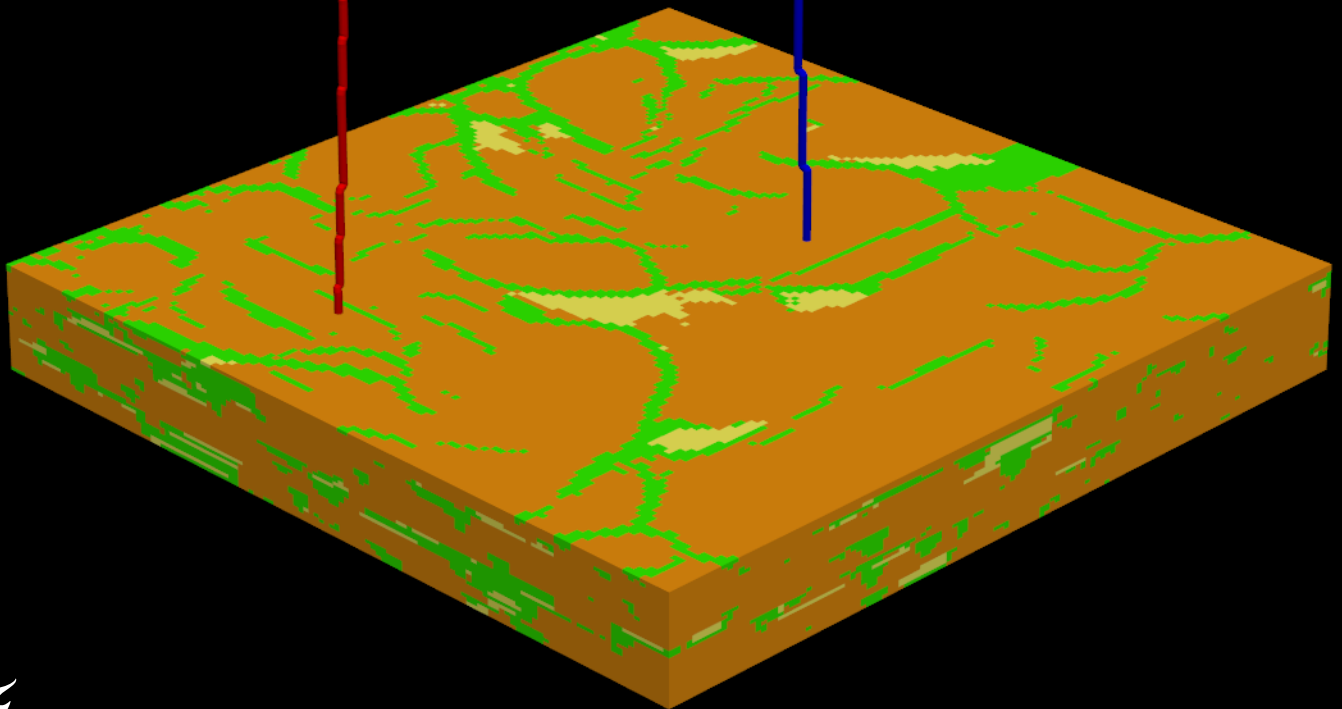


Evaluating the applicability of Generative Adversarial networks for geological facies modelling

A case study on the Delft Geothermal Reservoir

M.D. Torres Ruhe



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by

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Cover: 3D GAN realisation of the Delft Geothermal reservoir with the producer(red) and injector(blue) well trajectories. Assessment committee:

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An electronic version of this thesis is available at <http://repository.tudelft.nl/>. The code repository is

available at: <https://github.com/Matglassaa/Thesis-project-DGM>

Abstract

Traditional 3D geomodelling techniques face a significant compromise between computational efficiency and geological precision. While traditional methods struggle to simultaneously achieve high visual realism and adapt to hard well data, Deep Learning (DL) algorithms offer rapid sampling capabilities well-suited for large-scale uncertainty quantification. This research assesses the applicability of a Generative Adversarial Network (GAN), specifically a modified Fluvial Generative Adversarial Network (FluvGAN) architecture, to generate 3D realisations of the real-world Delft Sandstone (DSST) geothermal reservoir while conditioned to sparse real-world data. Robust synthetic training datasets were curated using the process-based stochastic modelling software FLUMY.

The results demonstrate that while a properly tuned 3-class GAN model successfully captures macro-scale 3D spatial dependencies and replicates dominant volumetric proportions, its performance is heavily bound by hyperparameter configurations. Stable adversarial dynamics were achieved by expanding network depth via double residual blocks and skip connections, combined with targeted R1 gradient regularization to preserve high-frequency spatial gradients. Expanding to a 9-class high-fidelity model or a model trained on a multi-prior dataset triggered severe optimization bottlenecks and ensemble mode collapse due to extreme categorical class imbalances within the training data.

To bridge this to practical field application, a two-step inversion workflow combining latent space optimization with pivotal tuning was deployed to anchor the synthetic realisations to hard physical constraints from the DEL-GT-01 and DEL-GT-02-S2 vertical well trajectories. This approach successfully matched the asymmetric local observations, yielding localized accuracy scores ranging from 0.78 to 0.92. However, true geological plausibility remains bottlenecked by topological artifacts—such as broken mudplugs and isolated voxels—in statistically suppressed minority facies, alongside a deflated spatial variance across the realization ensemble. This study concludes that while GANs offer an efficient pathway for ensemble-based uncertainty quantification, successful full-field subsurface deployment depends on advanced loss formulations and architectural upscaling to resolve fine-scale topological features without sacrificing stochastic diversity.

Acknowledgements

I would like to thank...

List of Abbreviations

DGM Deep Generative Modelling.

VAE Variational Auto Encoder.

GAN Generative Adversarial Network.

TUD Technische Universiteit Delft.

TvD True Vertical Depth.

DSST Delft Sandstone.

WNB West Netherlands Basin.

ANN Artificial Neural Network.

CNN Convolutional Neural Network.

DL Deep Learning.

FluvGAN Fluvial Generative Adversarial Network.

JSD Jensen-Shannon Divergence.

SWD Sliced Wasserstein Distance.

MDS Multi-Dimensional Scaling.

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1 Introduction

Sedimentary deposits are a vital commodity in modern society due to the various resources that can be extracted from them. These resources include minerals, groundwater, geothermal heat, sands and hydrocarbons (**morris2007genesis**; **donselaar2017relation**; **donselaar2015reservoir**; **brekke2001sedimenta**). Moreover, recent studies have investigated storage possibilities for carbon dioxide and hydrogen in this subsurface environment (**zhou2020dynamic**; **heinemann2021enabling**). Beyond the increase in subsurface applications, the world also deals with an increasing population and energy demand. Consequently, the demand for resources hosted within fluvial sedimentary deposits is expected to increase while the distribution of the commodities themselves becomes sparser and of lower quality (**spittler2020modelling**; **calvo2016decreasing**). This highlights the need for a deeper understanding of subsurface environments to enable more precise resource management which can be achieved through various methods such as: data gathering surveys, geological evolution and constructing high-quality 3D geomodels.

3D geomodels are traditionally constructed by engineers and geoscientists to predict reservoir properties and resource distribution (**song2022review**; **royer20153d**). The voxels (i.e. cells) in these models may be populated by continuous or categorical properties: grain size, porosity & permeability values or facies. The predictions are fundamentally based on a geological conceptual model which acts as a hypothesis regarding the most likely rock mass configuration, built entirely from sparse field data (**koltermann1996heterogeneity**). Data sparsity is an unavoidable reality of subsurface exploration, resulting from the exceptionally high costs and extended runtimes required for physical data acquisition. Therefore, engineers develop a working understanding of the subsurface based on limited information. This reliance on limited information drives the continuous need to improve geomodelling techniques. Such improvements aim to computationally bridge the gap between isolated data points and realistic, continuous geological representations.

1.1. Problem definition

Despite ongoing advancements in data collection, data scarcity dictates that functional 3D geomodels must rely heavily on subjective interpretations of the underlying geological conceptual model. Transitioning from these interpretive hypotheses into a digital framework requires applying specific mathematical and spatial algorithms thus introducing uncertainty at multiple stages, **stemming** from both the sparse data inputs and the modelling steps themselves (**hoyer2024evaluating**). As a result, precisely quantifying uncertainty is crucial for effective decision-making and can be accomplished with various techniques (e.g. **Monte Carlo Simulations (MCSs)**), generally asking for the creation of a large collection of probable 3D geomodels. However, creating this extensive ensemble reveals a core drawback in conventional modelling techniques: the significant compromise between computational efficiency and geological precision.

Figure 1.1 illustrates the trade-off between generating geologic realistic samples versus the ability to condition to hard well data. Where process- and rule- based methods excel at mimicking depositional processes, their forward-simulating nature makes it hard to inversely fit them to well-data. Pixel-based methods on the other hand offer the contrary: being able to fit easily to hard well data but lacking geologic realism. Finally, object base methods were introduced as a “best of both world’s” solution, placing geologic objects that fit the hard well data. However, due to the fact that these approaches are often limited to simple statistical distributions or strict object-oriented rules (**rongier2025atowards**), they **find it challenging** to replicate the intricate, non-linear architectural relationships present in natural sedimentary formations (**sun2023geological**).

As a result, existing workflows compel engineers to decide between models that can quickly produce a high number of samples while adhering to well data but do possess limited geological realism, **or geologically realistic models that require substantial effort to produce in large quantities** (**ENEMARK2019310**).

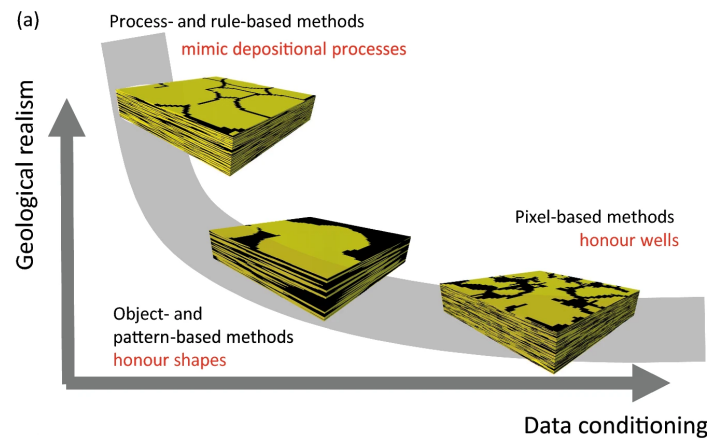


Figure 1.1: Simple overview of different geological modelling techniques compared to their ability to generate geological plausible data and honour well data (**manzocchi2026general**). Where process- and rule- based methods are great at generating geological plausible samples but struggle with honouring to well data, pixel-based methods are opposite in nature. Object based models offer a best of both worlds solution.

This bottleneck highlights a significant challenge in contemporary subsurface characterization. A significant need exists for a modelling approach that can thoroughly incorporate complex geological prior knowledge while also ensuring the fast generation rates necessary for ensemble-based uncertainty quantification. As a result, the question arises how the existing modelling process can be modified to ensure that:

- 1) Previous geological insights and structural information is better employed.
- 2) Uncertainty in the produced models is quantified more precisely and efficiently.

To address the need for both geological realism and rapid ensemble generation, recent research has proposed advanced **Machine Learning (ML)** algorithms as a highly effective solution. These algorithms, specifically the modern subset known as **DL**, possess the ability to learn complex, non-linear spatial relationships directly from data. By learning these intricate relationships, **DL** can accurately capture the geological complexity and non-stationarity that traditional methods often miss (**rongier2025atowards**). Furthermore, once such a model is fully trained and validated, the generative subset of **DL** models can generate a vast ensemble of 3D realisations at a fraction of the time compared to slower process-based models. This rapid generation directly facilitates the large-scale uncertainty quantification demanded by modern subsurface engineering.

1.2. Research gap and opportunity

Recent advances in 3D **Deep Generative Modelling (DGM)** have introduced various architectures for geomodelling, including **Variational Auto Encoders (VAEs)**, **Latent Diffusion Models (LDMs)**, and **GANs** (**laloy2017inver**, **bhavsar2024stable**; **song2023gansim**; **federico2025three**). Although **LDMs** currently lead in precise hard data conditioning, their iterative denoising process inherently bottlenecks inference speed. **GANs**, conversely, provide near-instantaneous sampling capabilities that are ideally suited for large-scale Monte Carlo uncertainty quantification. Despite known training challenges such as mode collapse, **GANs** are selected for this study due to their computational efficiency and established history in spatial generation. Training these models, however, requires an extensive volume of high-quality data to act as the geological prior. To overcome the natural scarcity of real subsurface observations, process-based forward modelling is utilized to generate robust synthetic training datasets.

Utilizing these synthetic datasets allows **GANs** to effectively learn geological priors; however, a critical research gap remains regarding their ultimate deployment. While **DGM** excels in theoretical environments, recent literature emphasizes an urgent need to transition from idealized, synthetic test cases to practical field applications. As **rongier2025atowards** **<empty citation>** stated, “We now need to move towards more detailed studies on larger case studies closer to field applications to further assess how the whole workflow performs”. Similarly, **federico2025three** **<empty citation>** noted that the

application (and extension as required) of the overall methodology to real cases should be performed. Addressing this explicit mandate forms the core opportunity of this research. Consequently, the primary goal of this project is to bridge the divide between synthetic training environments and practical field operations by assessing the performance of DGM in a more complex, real-world application.

Testing these models in a real-world application requires targeting an active subsurface system, a role fulfilled in this study by utilizing data from the DSST reservoir. This reservoir target must be characterized using a simple geological conceptual model combined with sparse well data. Such sparse well data will serve as the hard conditioning constraints to evaluate how the generative model behaves under tight physical limitations. These limitations, and the resulting fluctuations in uncertainty, will be analyzed by testing the workflow across multiple distinct geological priors. Investigating the interplay between varied priors and sparse conditioning constraints ultimately establishes the framework for the specific objectives of this study.

1.3. Objective and research questions

With the research gaps in mind, the main goal of this thesis will be to assess the applicability of using a GAN to generate realisations of a real-world reservoir by conditioning it to well data. Based on the outcome of this thesis, new insight can be given into the workflow needed to approach applying and evaluating GANs for generating subsurface reservoir realisations. Hence, the main research question of this work is: *To what extent can GANs be used to create geologically plausible realisations of the DSST reservoir while conditioned to sparse real-world data?* To help answer this question, several sub-research questions have been formulated:

1. How should the DSST's geology be parameterized in FLUMY to produce a training dataset sufficiently diverse for DGM modelling?
2. Which combination of hyperparameters yields a stable training process?
3. To what degree is the model able to reproduce the complexity inherent in the training data?
4. To what degree does the model generate realisations that are successfully conditioned to well data?
5. How different do models trained on different geological priors react when sparse data is introduced?

1.4. Thesis outline

The remainder of this thesis is structured to systematically address these research objectives. Chapter 2 establishes the theoretical foundation by reviewing the principles of geological modelling, the mechanics of deep generative models, and the specific geological setting of the DSST Member. Building upon this theoretical background, Chapter 3 outlines the core workflow, describing the generation of synthetic training datasets using FLUMY and detailing the specific GAN architecture deployed for the reservoir modelling tasks. The empirical outcomes of this workflow are presented in Chapter 4, which evaluates the training process and analyzes the structural and spatial realism of the generated realisations. These results are critically examined in Chapter 5, which contextualizes the findings, addresses the inherent limitations of the proposed approach, and highlights avenues for future research. Finally, Chapter 6 summarizes the main contributions of this thesis and provides definitive answers to the central research questions.

2 Background

This chapter establishes the theoretical and geological foundations and reasoning upon which this research is built. It provides a short overview of the geological setting of the Delft reservoir and a literature review of modern geomodelling and ML techniques. Section 2.1 introduces the geological context of the Delft reservoir, detailing its depositional environment and current conceptual model. Section 2.2 explores the state-of-the-art in geological modelling, comparing traditional techniques and their respective limitations. Finally, Section 2.3 provides a general overview of ML and DL principles, focusing on their historical evolution and current application to subsurface characterization. This balanced overview is intended to provide both geological and computer science readers with the necessary context to understand the research methodology and results presented in subsequent chapters.

2.1. The Delft Geothermal Reservoir

The data utilized in this study is sourced from the Delft geothermal reservoir, situated beneath the Technische Universiteit Delft (TUD) Campus in the Netherlands. As of February 2026, a geothermal doublet is actively producing heat from this reservoir (beerda2026geothermie). A doublet system consists of a two-well setup: a producer well extracts warm formation water, and an injector well returns the cooled water to the reservoir. At the Delft site, the producer well (DEL-GT-01) was drilled to a True Vertical Depth (TvD) of approximately 2,337 m, while the injector well (DEL-GT-02-S2) reached a TvD of approximately 2,100 m (vardon2024endofwell). The project serves both commercial and research objectives, acquiring extensive geophysical logs and core data, supplemented by surface and downhole monitoring instrumentation (vardon2020geothermal; vardon2024endofwell). Despite the high data density, the cored section is restricted to the first 80 m of the DEL-GT-01 well (approximately one-fourth of the reservoir thickness), while DEL-GT-02-S2 contains no cored sections. The primary target is the DSST, a member of the Lower Cretaceous Nieuwerkerk Formation in the West Netherlands Basin (WNB) (donselaar2015reservoir).

2.1.1. Geology of the Delft Reservoir

The WNB is a 60 km wide transtensional basin in the southwest of the Netherlands, characterized by NW-SE trending structural highs and depressions (ziegler1990geological; donselaar2015reservoir). The WNB entered an active rifting stage during the Middle Jurassic (bodenhausen1981habitat; den1996fluviomarine; willems2020geology), leading to the formation of several half-grabens filled with terrestrial sediments sourced from neighboring regions (den1996fluviomarine). The progressive filling of these half-grabens coincided with a change in the depositional environment as a transgressive shoreline moved the system from fluvial to marine. The Nieuwerkerk Formation comprises three members: the Alblasserdam, the DSST, and the Rodenrijs Claystone.

The Alblasserdam member was deposited in a fluvial environment characterized by stacked channels and floodplain deposits (devault2002tectonostratigraphy). The DSST directly overlays the Alblasserdam and consists of thick, stacked fluvial-deltaic channel sandstones with high net-to-gross values, oriented NW-SE (willems2020geology; mijnlieff2020introduction; bruining2023sedimentological; groenenberg2010targeting). The DSST is in turn overlain by the Rodenrijs Claystone, an organic-rich lagoonal deposit (willems2020geology; devault2002tectonostratigraphy; bruining2023sedimentological).

Based on the sedimentological analysis of the DEL-GT-01 core and DEL-GT-02-S2 sidewall samples, three primary Facies Associations (FAs) have been identified (bruining2023sedimentological):

- **FA1: Fluvial and distributary channel deposits.** Primarily light grey, fine- to medium-grained sandstones featuring low-angle cross-bedding and ripple lamination, with unit thicknesses ranging from 5 to 80 m.
- **FA2: Crevasse splays and levee deposits.** Composed of moderately sorted, very fine- to medium-

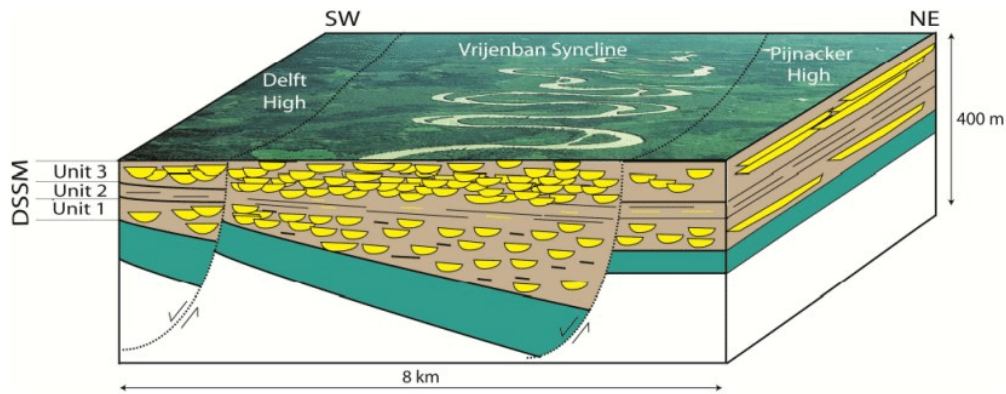


Figure 2.1: Depositional models for the DSST Member with three identified units. Accommodation space is increasing towards the NE ([donselaar2015reservoir](#))

grained sandstones interbedded with dark grey siltstones, with thicknesses ranging from 1 to 5 m.

- **FA3: Overbank deposits and paleosols.** Consists of dark grey siltstones, claystones, and very fine-grained sand deposited in low-energy environments. This association is characterized by abundant organic material, slickensides, and siderite nodules.

The specific depositional environment of the DSST remains a subject of scientific discussion. It has been attributed to both fluvial-meandering ([donselaar2015reservoir](#); [vondrak2018reservoir](#); [willems2020geology](#)) and fluvio-deltaic systems ([van1993stratigraphic](#); [weert2024multiple](#); [bruining2023sedimentological](#)). Recent analysis by [bruining2023sedimentological](#) suggests that the cored section represents a deltaic environment. The observed FAs can be attributed to a upper to lower plain delta with multi-stacked fluvial/distributary channels, single channels, crevasse splays, levees and overbank fines.

2.1.2. Deltaic Depositional Environment

Deltaic environments form at the land-sea interface where river systems supply clastic sediments. Delta morphology is governed by the relative influence of fluvial, wave, and tidal processes ([kolb1966depositional](#)). The subaerial delta plain is typically divided into two domains ([kolb1966depositional](#); [heldreich2017challenges](#)):

- **Upper Delta Plain:** Lies above the limit of saltwater intrusion. Sedimentation is driven by distributary channel migration, overbank flooding, and crevasse splaying. Environments include meandering and braided channels, lacustrine fills, and floodplains.
- **Lower Delta Plain:** Subject to river-marine interaction and tidal influence. Channels are more numerous and exhibit bifurcating or anastomosing patterns. Interdistributary bays, bay fills (crevasse splays), and marshes dominate the landscape.

In the DSST, FA1 features organic material and siltstones indicating slackwater periods, while the lack of bioturbation suggests high sedimentation rates. The fine-grained nature of FA2 and FA3 reflects more distal positions on the delta plain. Siderite nodules in FA3 indicate anoxic, waterlogged conditions ([samuel1959some](#)), while soft-sediment deformation suggests rapid loading of mudstones over saturated sands. These features collectively indicate that FA3 represents deposition within a delta plain, likely in an interdistributary bay or a pro-delta environment.

2.2. Geological Modelling

The primary objective of 3D reservoir modelling is to provide a representative volume of the subsurface for flow simulation and decision support. These models are dynamic, requiring updates as new data becomes available to refine predictions over the reservoir's lifetime ([emerick2025ensemble](#); [ringrose2016reservoir](#)). Existing techniques are generally categorized into three types ([koltermann1996heterogeneous](#)):

- **Pixel-based Models:** Use spatial statistics (e.g., variograms) for cell-by-cell property assignment. They are efficient and honor dense well data but often fail to capture complex connectivity,

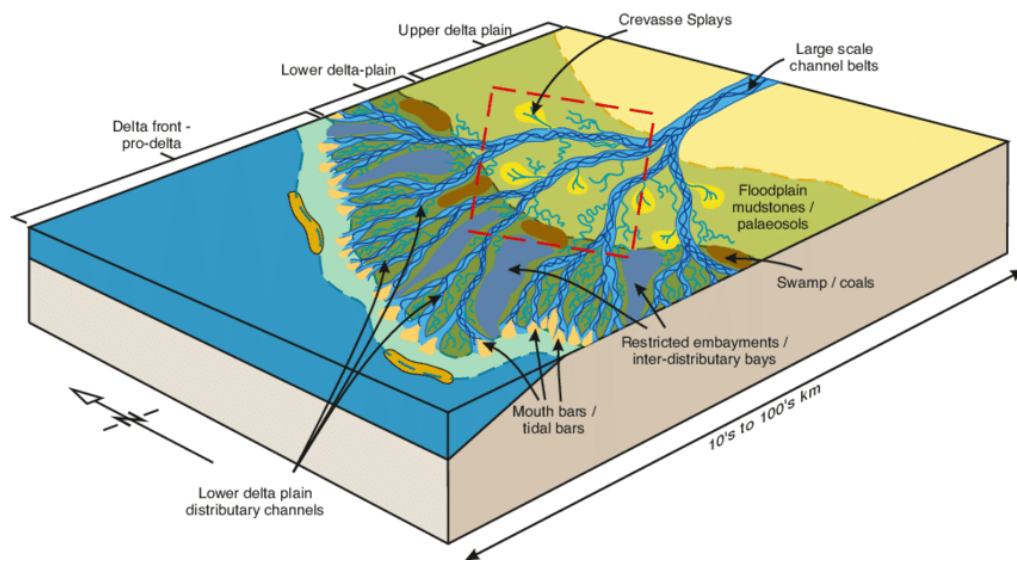


Figure 2.2: Depositional model of the Mungaroo Formation, illustrating the geometries and domains of a fluvial-deltaic system (heldreich2017challenges).

resulting in pixelated realisations.

- **Object-based Models:** Directly place geometric features (e.g., channels) based on geological rules. They offer high visual realism but struggle to honor dense or closely spaced well data.
- **Process-based Models:** Simulate the processes behind sediment transport and deposition. They provide the highest degree of realism but are computationally expensive and difficult to condition to specific hard data like well logs.

Attribute	Geostatistical	Object-Based	Process-Based
Geological Prior	Variograms, training images.	Geometric shapes and rules.	Physical boundary conditions.
Realism	Low to Moderate.	Moderate to High.	Very High.
Key Advantages	Computational speed, well conditioning.	Realistic connectivity.	Physical consistency.
Key Disadvantages	Lack of geological realistic shapes.	Difficult conditioning.	High computational cost.

Table 2.1: Comparison of spatial modelling techniques in geomodelling.

While traditional pixel-based geomodelling relies heavily on two-point spatial statistics, such as variograms, to distribute properties, these methods often fail to capture the complex, multipoint connectivity inherent in natural geological systems. ML introduces a paradigm shift by moving away from user-defined spatial rules. Instead of requiring an engineer to pre-define the shape of a channel or the range of a variogram, ML algorithms learn these intricate, non-linear spatial relationships directly from data. This makes ML exceptionally well-suited for geomodelling, where the relationships between sparse hard data (e.g., well logs) and the broader geological architecture are too complex to be captured by standard statistical interpolation.

2.3. ML

ML involves the development of algorithms that learn patterns from data to perform specific tasks without explicit programming (goodfellow-et-al-2016). This foundational concept of learning without explicit programming has evolved significantly since samuel1959some<empty citation> intro-

duced the term, expanding into a wide array of computational techniques. These diverse techniques are fundamentally categorized based on their training approach into two primary domains:

- **Supervised Learning:** The algorithm is trained on a labeled dataset containing explicit input-output pairs.
- **Unsupervised Learning:** The algorithm identifies hidden patterns within unlabeled data without prior knowledge of the desired output.

Within these two broad categories, algorithms are deployed to solve common tasks such as classification, regression, clustering, and generation. Applying these generative tasks to the subsurface highlights a severe limitation in traditional pixel-based geomodelling, which relies heavily on simple two-point spatial statistics (e.g., variograms) to distribute properties. Because these simple statistics often fail to capture the complex, multipoint connectivity inherent in natural geological systems, ML introduces a necessary paradigm shift. This shift moves the modelling workflow away from strict, user-defined spatial rules. Instead of requiring an engineer to pre-define the range of a variogram, ML algorithms learn intricate, non-linear spatial relationships directly from the data. Learning directly from data makes ML exceptionally well-suited for geomodelling, where the relationships between sparse hard data (e.g., well logs) and the broader geological architecture are simply too complex to be captured by standard statistical interpolation. The capacity to uncover such complex, non-linear relationships in vast datasets is largely driven by modern Artificial Neural Networks (ANNs) (lecun2015deep). Inspired by the biological structure of the brain, these ANNs consist of interconnected artificial neurons that process information across multiple computational layers (goodfellow-et-al-2016).

While ANNs and traditional ML algorithms excel at analyzing tabular petrophysical data for various subsurface tasks, they inherently lack actual geological reasoning skills. Without such reasoning to naturally make sense of areas with low data quality, these algorithms are restricted to inferring abstract features directly from their training data (schaaf2019quantification). Inferring these features in standard ML workflows necessitates manual feature extraction, a process where engineers must explicitly define the relevant geological parameters prior to training. This strict reliance on manual feature extraction creates a critical limitation when attempting to scale these algorithms to 3D geomodels. Within these high-resolution 3D geomodels, which typically contain millions of voxels, manually defining multi-scale spatial features ultimately becomes computationally impractical.

DL overcomes this limitation through automatic feature extraction. By utilizing multi-layered networks, DL architectures natively process high-dimensional spatial grids, bypassing the manual extraction bottleneck. For subsurface characterization, this means a DL model can ingest raw 3D conceptual models or process-based simulations and autonomously identify the underlying geological rules. This capability to scale with high-dimensional data makes DL the premier mathematical framework for modern reservoir modelling.

2.3.1. DL and Inductive Biases

DL is a subset of ML focusing on multi-layered neural networks that represent data with multiple levels of abstraction (goodfellow-et-al-2016; lecun2015deep). Achieving these levels of abstraction in high-dimensional spaces introduces a fundamental challenge known as the “curse of dimensionality.” This curse of dimensionality describes the exponential increase in data required as the spatial dimensions grow, making high-dimensional estimation computationally challenging. To overcome this computational challenge and help models generalize from limited data, engineers implement specific architectural choices known as inductive biases (prince2023understanding). A premier example of such an inductive bias is the foundational architecture used for processing spatially structured data: the Convolutional Neural Network (CNN).

CNNs leverage spatial inductive bias to become highly effective for 2D and 3D geomodelling tasks. Executing these spatial tasks relies on a core architectural innovation known as the convolutional layer. This convolutional layer replaces the fully connected neurons of a standard ANN with a localized set of learnable filters, commonly referred to as kernels. These multi-dimensional kernels operate by systematically sliding across the input data in discrete increments dictated by a predefined step size, or stride. This sliding stride is frequently combined with padding, a technique that adds artificial zero-value borders to the input to ensure the entire spatial region is comprehensively analyzed. Analyzing

this spatial region involves computing a local dot product between the kernel weights and the input values, a mathematical operation visually illustrated in Figure 2.3.

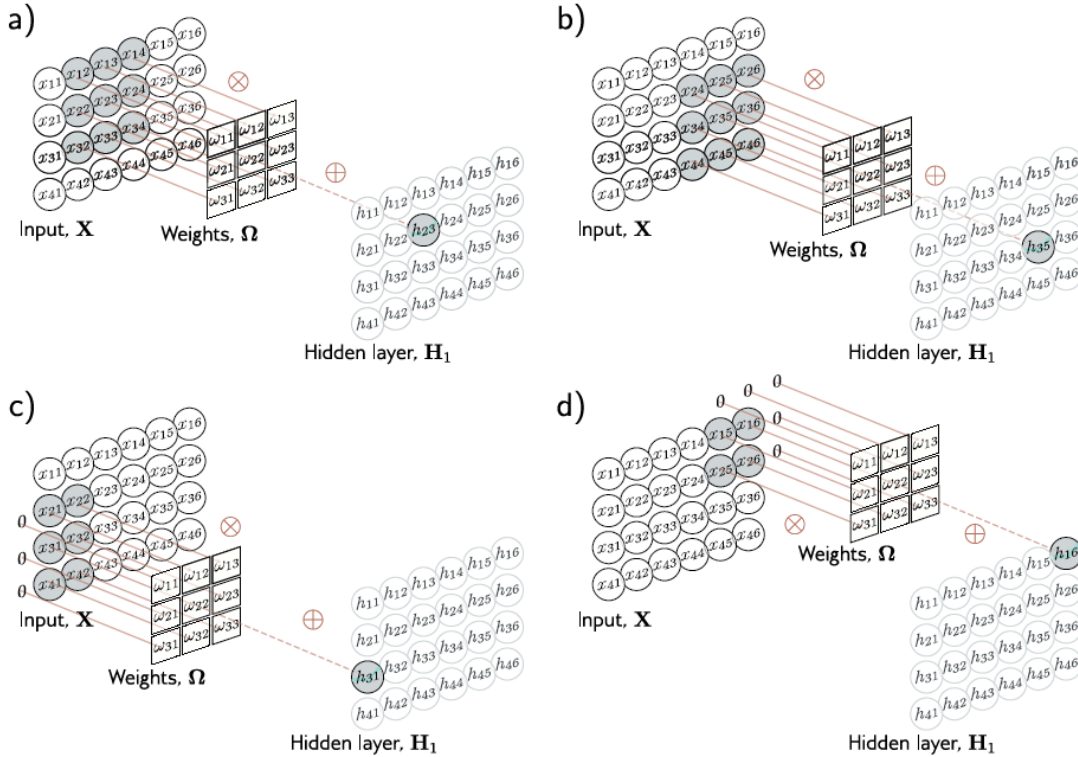


Figure 2.3: Example of a 2D convolutional layer with kernel size of 3x3 and a padding of 1. Each output h_{ij} is composed of a weighted sum of the 3x3 nearest inputs on the previous layer (**prince2023understanding**). a) Here, the output h_{23} (shaded output) is a weighted sum of the nine positions from x_{12} to x_{34} (shaded inputs). b) Different outputs are computed by translating the kernel across the image grid in two dimensions. c–d) With zero-padding, positions beyond the image's edge are considered to be zero.

The mathematical result of this operation is a new, filtered spatial representation known as a feature map. This feature map is typically downsampled by a subsequent pooling layer, which condenses the spatial dimensions while preserving the most prominent information. Preserving this information across a continuous stack of convolutional and pooling layers allows the network to progressively map low-dimensional input data into high-dimensional, abstract features. **At the base of this feature mapping process, the initial layers capture simple, low-level spatial patterns, such as discrete edges or sharp facies boundaries.** By mathematically combining these simple low-level edges in deeper layers, the network learns to recognise highly complex, high-level geological architectures, such as the overall sinuosity of a fluvial channel. Learning these complex high-level architectures is computationally efficient because the exact same kernel is applied universally across the entire grid, creating an inherent advantage known as translation invariance. This translation invariance ensures that if the network learns to identify a channel structure in one corner of the reservoir, it can accurately recognise that same structure anywhere else in the model, drastically reducing computational overhead and preventing overfitting. Further preventing this overfitting requires another critical component of the DL workflow: the strict use of validation and testing protocols. These validation protocols ensure that the hierarchical patterns learned by the network are robust and safely applicable to unseen data. Applying these robust, learned spatial patterns to unseen subsurface data forms the fundamental basis of modern DL-driven geomodelling workflows.

2.3.2. Geomodelling with DL

Modern DL-driven geomodelling workflows have evolved significantly, transitioning from simple pixel-based interpolations to the generation of highly sophisticated spatial representations. Generating these

sophisticated representations relies on the ability of DL to directly learn complex geological features from large datasets or process-based simulations. Learning directly from such simulations bridges the historical gap between simplistic descriptive statistics and true geological realism (**laloy2017inversion**). To achieve this realism, CNNs have proven particularly successful in capturing the complex, non-stationary architectures of fluvial channels and facies distributions (**sun2023geological**). Capturing these intricate architectures enables the rapid generation of stochastic realisations that successfully honor both the global geological concept and local, sparse conditioning data. Honoring these dual constraints while generating massive ensembles of stochastic realisations fundamentally enhances subsurface uncertainty characterization (**song2023gansim**). This unique capacity for rapid, constrained generation is primarily driven by specialized neural network frameworks, most notably generative architectures.

These generative architectures that are using CNNs, are best known for their ability to generate realistic images. The most-well known models include VAEs, GANs and LDMs. Though they are all different in terms of architecture, some core concepts remain similar. Generative models are designed to learn and approximate the complex joint probability distribution of the training data whereby elements are generated based on previously generated values (**he2025generative**). Moreover, these models use a latent space, designed to either store the encoded training data (VAEs) or be used to scale to an image through various upsampling steps (GANs and LDMs) (**prince2023understanding**).

2.3.3. GANs

Generative architectures, specifically GANs, achieve complex spatial generation through a competitive framework between two distinct neural networks (**goodfellow2014generative**). These two networks are defined as a Generator (**G**) and a Discriminator (**D**). The generator is tasked with creating synthetic spatial samples from random noise, while the discriminator is simultaneously tasked with distinguishing these synthetic samples from actual real-world training data. The continuous competition to outsmart one another drives a minimax game, which is mathematically expressed as:

$$\min_{\mathbf{G}} \max_{\mathbf{D}} V(\mathbf{D}, \mathbf{G}) = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}(\mathbf{x})} [\log \mathbf{D}(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_{\mathbf{z}}(\mathbf{z})} [\log(1 - \mathbf{D}(\mathbf{G}(\mathbf{z})))] \quad (2.1)$$

In this equation, $\mathbf{x}$ represents the real data, and $\mathbf{z}$ represents a random latent vector sampled from a prior distribution $p_{\mathbf{z}}$. The first term represents the discriminator successfully recognizing real geological data, while the second term represents the discriminator successfully catching fake data generated by $\mathbf{G}(\mathbf{z})$. During training, the discriminator seeks to maximize this overall value function to perfect its detection abilities, while the generator seeks to minimize it to successfully fool the discriminator. An overview of these steps, describing a single iteration of the model, is shown in figure 2.4. By successfully fooling the discriminator, the generator ultimately learns to produce high-fidelity geological realisations without requiring any explicit, hard-coded spatial rules.

GAN Inversion

Figure 2.4 illustrates the generation of high-fidelity, unconstrained realisations, which is a powerful capability; however, practical reservoir modelling requires these models to honor sparse physical constraints like hard well data. Hard well data constraints are typically honored through optimization techniques (**rongier2025btowards**; **bhavsar2024stable**; **chan2019parametric**). Several techniques have been proposed including applying inference networks (**bhavsar2024stable**; **chan2019parametric**), Gaussian Mixture Modelling (GMM) (**bhavsar2024stable**), latent space optimization and inversion (**rongier2025btowards**). Where inference and GMM (ADD EXPLANATION ON THESE TECHNIQUES) GAN inversion operates strictly within the model's latent space after the training phase is complete. Completing the training phase allows the algorithm to iteratively search and optimize this latent space to identify a specific latent vector $\mathbf{z}$. This specific latent vector $\mathbf{z}$ generates a 3D volume perfectly matching the sparse well data. Matching the sparse well data securely conditions the generative model to real-world subsurface observations. Moreover, a second proposed technique by **rongier2025btowards** is pivotal tuning, a technique originally designed to fine-tune the generator to comply a set latent space (**roich2022pivotal**).

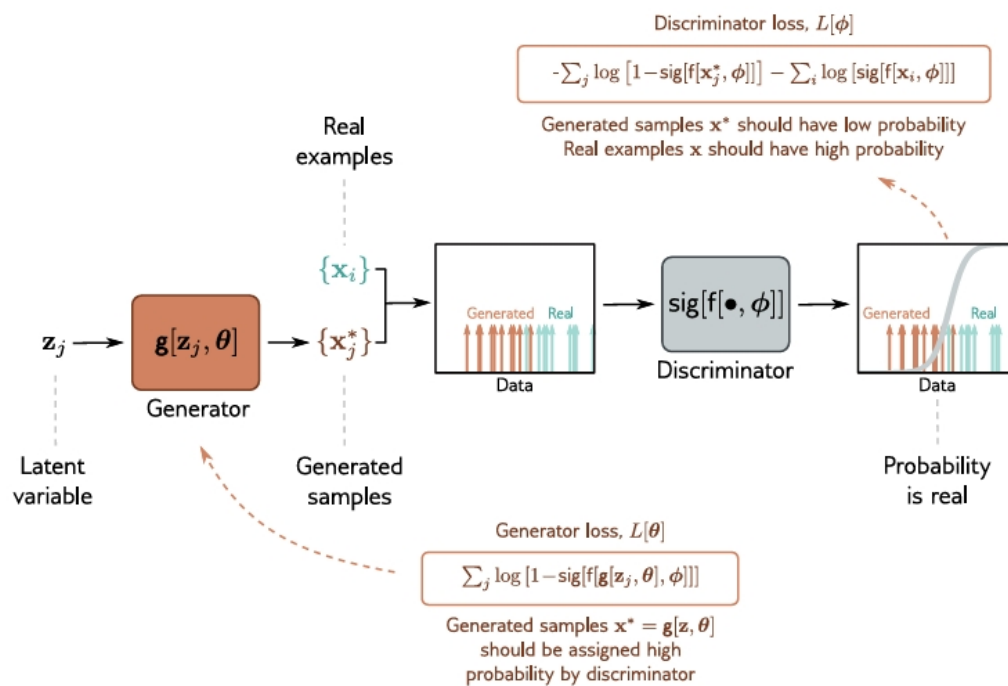


Figure 2.4: Basic overview of a standrd GAN architecture with a Generator, constructing fake samples from a random latent variable z_j and a Discriminator deciding which of the two samples is the real sample (**prince2023understanding**).

3 Methodology

This chapter describes the project workflow, beginning with the generation of 3D synthetic datasets used for training, validating, and testing the models (Section 3.1). Section 3.2 details the GAN framework, including the network architectures, the facies representation strategy, and the loss functions employed during training. Finally, Section 3.3 introduces the metrics and simulation methods used to evaluate the geological realism and practical utility of the generated realisations.

The primary real-world constraints for this project are derived from the physical well logs of the DEL-GT-01 and DEL-GT-02-S2 geothermal wells. bruining2023sedimentological<empty citation> interpreted the cored sections of these wells, translating the raw physical logs into distinct, combined FAs. The FAs along the wellbore are presented in figure B.1. The specific categorical nature of this well data serves as the link between the synthetic generative models and practical field application, driving two key methodological decisions in this project:

1. Because facies classifications denote discrete geological environments rather than continuous variables, they lack any numerical relationship (e.g., an integer index of 2 does not imply twice the magnitude or a higher value than an index of 1). Treating them as raw scalar values would introduce numerical dependency, incorrectly forcing the GAN to optimize for non-existent trends between distinct lithologies.
2. The FAs of these well-log interpretations, which group the complex subsurface into three primary associations, directly necessitate the subsequent grouping of the generated dataset(s) into upscaled 3-facies dataset(s).

By aligning the model’s categorical resolution with the exact structure of the available well data, it ensures that the generative framework can be practically conditioned on and evaluated against actual physical subsurface evidence.

3.1. Datasets

In order to test the applicability of GANs, new datasets had to be generated since training DL algorithms for 3D geomodelling requires a substantial volume of high-fidelity training data. The new datasets were generated using FLUMY (martin-hal-01313232), a process-based stochastic modelling software designed to reproduce meandering channelized reservoirs. Based on sedimentological rules, the software reproduces facies type, grain size and deposition time for each voxel in th 3D volume.

3.1.1. Dataset Descriptions

To test the GAN in different scenario’, three distinct geological conceptual models were curated to capture a range of environments for the DSST reservoir. The setting specific input parameters are shown in Table 3.1. The hypothesized depositional environments are split across three discrete suites, rang-

Parameter	Setting 1	Setting 2	Setting 3
Net-to-Gross (%)	67	67	67
Max. Channel Depth (m)	5	6	7
Max. Sand Body Extension	80	100	120

Table 3.1: FLUMY NEXUS (Non-EXpert User Settings) input parameters for the three training datasets.

ing from a Lower plain to Upper plain deltaic characteristics in terms of increasing channel depth and wider sand bodies. Beyond these constraints, all generated models share a uniform reservoir scale with a total grid resolution fixed at 32x128x128 voxels. In order to allow for well placement in the samples, the final dimension should encompass an area of at least 2000x2000m as the DEL-GT-01

and **DEL-GT-02-S2** have an 1500m spacing between them. Establishing this specific grid resolution combined with a standard voxel dimension of 20 m×20 m×1 m means that the total simulated domain covers a volume of 2560 m×2560 m×32 m. Every single voxel within this acute 32 m vertical interval is ultimately encoded with 1 of the 14 distinct facies (see B.1) available in FLUMY, of which only 9 facies actually occurred in the sample ensemble due to the specific parameters for each geological setting.

3.1.2. Sample preprocessing

To assess the capabilities of GANs for the Delft Geothermal Reservoir, the synthetic training datasets are prepared at two distinct categorical resolutions. These two distinct resolutions consist of a high-fidelity 9-facies dataset and an upscaled 3-facies dataset. The high-fidelity 9-facies dataset preserves the detailed depositional environments generated by FLUMY, serving as a baseline to evaluate the model's pure generative capacity and its ability to replicate highly complex spatial distributions. While evaluating this generative capacity is crucial, practical field application requires conditioning the model to actual subsurface observations. These actual subsurface observations, derived from the cored section of the **DEL-GT-01** well, are strictly classified into three primary FAs (Section 2.1.1). Because the hard well data only resolves to three distinct associations, the original FLUMY depositional environments must be grouped accordingly to create the upscaled 3-facies dataset. This grouping process aggregates the detailed environments into broader categories, as detailed in Table 3.2.

Value	Facies Key	Color	Description
1	Channel & Point Bar (FA1)		Grouped proximal deposits: Channel Lag (1), Point Bar (2), Sand Plug (3), and Crevasse Splay I (4).
2	Levee & Crevasse (FA2)		Grouped intermediate deposits: Crevasse Channel (5), Crevasse Splay II (6), and Levee (7).
3	Overbank Fines (FA3)		Grouped distal deposits: Overbank (8), Mud Plug (9), Hemipelagic Plug (10), Wetland/Paleosol (11), Draping (12), and Pelagic (13).

Table 3.2: Overview of the upscaled 3-facies grouping used for the generative modelling, aggregated from the original 13 FLUMY depositional environments (See Table ?? for the original facies).

Beyond this grouping process, the categorical nature of both the 3-facies and 9-facies datasets necessitates a final preprocessing step known as one-hot encoding (**goodfellow-et-al-2016**). One-hot encoding is critical because standard neural networks process continuous numerical gradients and might otherwise incorrectly interpret integer facies labels as having a mathematical relationship. This misinterpretation is prevented by transforming the original 3D volume into an expanded 4D tensor, where each class is assigned its own dedicated spatial channel. Each spatial channel indicates the presence or absence of a specific facies using values of 1 and -1 (See appendix in C.1).

3.2. Fluvgan Framework

3.2.1. Model Architecture

The GAN architecture used in this project is based on the **FluvGAN** model, introduced by **rongier2025a**fluvgan<empty citation>. Where **rongier2025a**towards<empty citation> used a 2-channel approach for their model, one for grain size and one for deposition time, the model used in this project comprises of either 3 or 9 channels, where each channel represents another facies based on the one-hot encoding strategy. The deposition time channel was not used in this project.

The architecture of the model is comprised of a generator (**G**) and a discriminator (**D**) built with 3D convolutional blocks. These blocks enable the generator to transform a, in this case, 100-dimensional latent vector $\mathbf{z}^*$ into a $C \times 32 \times 128 \times 128$ volume, where C refers to the channels. The volume begins as a $1024 \times 4 \times 4 \times 4$ feature map, created via a 3D transposed convolution, and is then expanded by five residual upscaling blocks. The final spatial resolution is processed through a Tanh activation, ensuring

the C facies channels are scaled from -1 to 1. Finally, an argmax function scales the one hot encoded tensor back to integers, ensuring that a $32 \times 128 \times 128$ volume inhabited by class numbers remains. Every generated voxel is then evaluated by the discriminator, acting as a binary classifier to reduce the $C \times 32 \times 128 \times 128$ input to a single scalar logit. This logit is produced through five residual downscaling blocks that iteratively reduce spatial resolution while increasing the channel count to 1024. A simple visual representation of the model can be seen in figure 3.1. In total, the generator involves roughly 21.0 million trainable parameters while the discriminator has about 7.6 million trainable parameters.

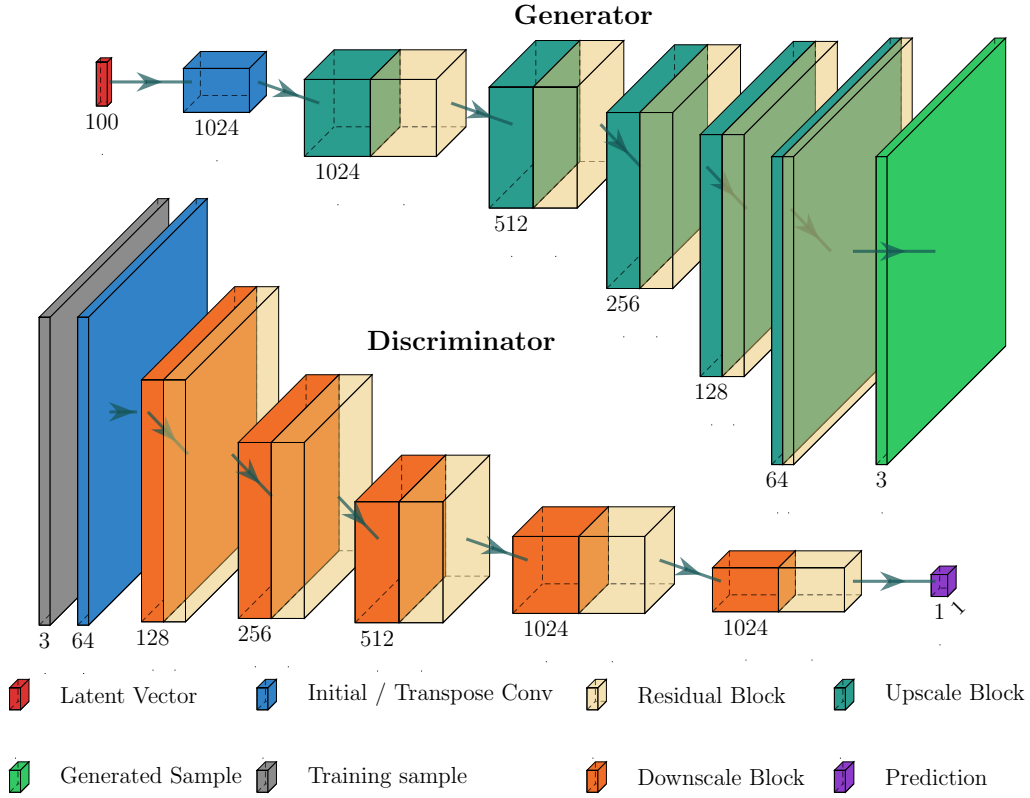


Figure 3.1: **Top)** The Generator initiates from a 100-dimensional latent vector and expands spatial volume through successive upscale and residual blocks. During this process, feature channel density is progressively reduced from 1024 down to 3 channels to construct the final generated volume. **Bottom)** The Discriminator mirrors this pathway by receiving a 3-channel sample (training or generated) and contracting spatial dimensions through downscale blocks. To retain high-level structural patterns during downsampling, feature channels are steadily increased from 3 up to 1024 before compressing into a single purple block representing the final prediction scalar.

During the adversarial training process, the weights of both the Generator (**G**) and the Discriminator (**D**) are iteratively updated according to an objective function. The goal of this function is to quantify adversarial performance: measuring how accurately the discriminator distinguishes authentic geological data from synthetic samples, and conversely, how effectively the generator produces samples that fool the discriminator. Since the adversarial loss functions are purely based on the discriminator's outputs, which acts as a binary classifier (a sample is 'real' or fake), the appropriate loss function is the standard Binary Cross-Entropy (BCE) loss. Whether the underlying data represents continuous properties or categorical one-hot encoded facies, the adversarial objective remains a binary classification problem at the discriminator level. Mathematically, evaluated across a batch of N samples, the standard BCE loss is defined as:

$$\mathcal{L}_{BCE} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(D(x_i)) + (1 - y_i) \log(1 - D(x_i))] \quad (3.1)$$

where y_i represents the binary domain label (real or fake), and $D(x_i)$ is the probability assigned by the discriminator.

During adversarial training, instability often manifests as exploding or saturating logarithmic losses. From an objective function perspective, these dynamics depend heavily on the discriminator’s output; if the discriminator becomes completely confident in its classifications, the gradients provided to the generator can vanish. Severe instability or sudden drops in generator loss occur when the generator successfully exploits a specific artifact or shortcut that completely fools the discriminator, leading to a loss of gradient diversity. Such uncalibrated training dynamics make the framework highly susceptible to severe mode collapse (**prince2023understanding**), where the generator’s capability is restricted, and it ceases to produce diverse geological outputs. Beyond the foundational loss formulation, the specific interplay of model architecture and optimization hyperparameters dictates the overall convergence. To evaluate the configurations that yield stable adversarial dynamics and realistic geological outputs, a hyperparameter tuning study was conducted.

Hyperparameters

Hyperparameter tuning involves testing various combinations of network settings to evaluate their impact on model performance. The primary objective is to identify an optimal configuration that minimizes the discrepancy between the generated models and the input data, while actively preserving spatial ensemble diversity. To achieve this, baseline parameters (such as the optimizer and batch size) were kept constant (see table 3.3), while a specific set of architectural and training hyperparameters were varied stepwise. The complete search space for this hyperparameter study is outlined in Table 3.4.

Hyperparameter	Fixed Value	Description
Optimizer	Adam ($\beta_1 = 0, \beta_2 = 0.999$)	Standard optimization algorithm.
Batch Size	64	Number of samples processed per iteration.
Kernel Size	$3 \times 3 \times 3$	Dimensions of the 3D Convolutional filters.

Table 3.3: Fixed baseline configurations for the Fluvgan architecture optimization study.

Hyperparameter	Search Space / Value	Description
Learning Rate G (α_G)	$10^{-3}, 10^{-4}$, Dynamic	Generator step size during gradient descent.
Learning Rate D (α_D)	$3 \times 10^{-3}, 10^{-2}$, Dynamic	Discriminator step size; testing a two-timescale update rule.
num iter D	1, 2	Number of consecutive Discriminator updates per Generator update.
Discriminator Penalty	None, 1, 16, 32	Gradient penalty applied to stabilize Discriminator confidence.
Double Conv Blocks	On, Off	Architectural toggle for adding secondary convolutional layers.
Double ResBlocks	On, Off	Architectural toggle for adding secondary residual blocks.
Training Samples	10,000, 20,000	Total volume of synthetic baseline data fed to the network.

Table 3.4: Hyperparameter search space for the Fluvgan architecture optimization study.

The Learning rates α_G and α_D dictate how magnitude of the network’s weights update while the num iter **D** refers to the discriminator update frequency compared to the generator. The Discriminator Penalty serves as a vital regularizer to constrain gradient norms, ensuring the discriminator’s confidence does not overpower the generator early in training. Lastly, the architectural toggles (Double Conv and Residual Blocks) paired with data scaling systematically isolate whether network capacity or dataset volume is primary in preserving the necessary spatial and structural diversity of the generated fluvial systems.

3.2.2. Optimization & Inversion

While the 9-facies models are utilized to evaluate unconstrained generative performance, the 3-facies models are deployed to test practical conditioning against the **DEL-GT-01** well data. To successfully condition this generative model to real-world observations, the inversion workflow (introduced by (rongier2025btowards)) relies on a two-step mechanism: Latent Space Optimization and Pivotal Tuning (roich2022pivotal).

First, Latent Space Optimization searches for the optimal random noise vector $\mathbf{z}$ that best approximates the well data while keeping the pre-trained generator’s weights frozen. This iterative search is driven by a composite loss function designed to satisfy two simultaneous objectives: honouring the hard conditioning well data and preserving global geological realism. The total optimization loss ($\mathcal{L}_{total}$) combines a context loss ($\mathcal{L}_{context}$) representing the well mismatch, and a prior loss ($\mathcal{L}_{prior}$) evaluated using the Discriminator’s Binary Cross-Entropy (BCE) metric:

$$\mathcal{L}_{total} = \mathcal{L}_{context} + \lambda \mathcal{L}_{prior} \quad (3.2)$$

Here, $\mathcal{L}_{context}$ explicitly penalizes deviations between the generator’s output channels and the true one-hot encoded well observations, utilizing distance-based thresholding to localize the penalty. Simultaneously, $\mathcal{L}_{prior}$ ensures the generated sample remains mathematically indistinguishable from real data. By feeding the updated generation to the frozen discriminator and calculating the BCE loss, it prevents the latent vector from degrading into non-geological noise. In this specific implementation, the prior BCE loss is heavily weighted with a multiplier of $\lambda = 10.0$ to guarantee structural integrity during the gradient updates.

Because this latent space optimization often provides a close structural approximation but rarely a perfect voxel-by-voxel match, Pivotal Tuning provides a second phase to resolve this. This second phase uses the optimized noise vector as a static anchor and temporarily unfreezes the generator’s internal weights to make microscopic adjustments. This forcefully aligns the localized output with the exact well data, while regularization prevents the collapse of broader geological realism. (A detailed mathematical formulation of this inversion process is provided in Appendix D.1).

3.3. Evaluation Metrics

Evaluating the performance of generative models on 3D spatial grids presents significant challenges, as standard metrics like Binary Cross-Entropy (BCE) or direct voxel-by-voxel assessments fail on unconditioned models. Therefore, the evaluation framework relies on metrics that capture complex geological structures and spatial uncertainty across an ensemble of generated samples.

To evaluate how closely the generated realisations resemble the training data’s global morphological structures, we employ the **Multi-Scale Sliced Wasserstein Distance (MS-SWD)** (MSSWD). Based on the optimal transport principles of the Wasserstein distance (panaretos2019statistical), which was highlighted for geological 3D structures by rongier2025atowards<empty citation>. MSSWD evaluates structural similarity without requiring rigid pixel alignment. By projecting 3D spatial distributions onto 1D linear slices over multiple downsampled resolution scales and averaging the results, MSSWD efficiently bypasses the computational bottleneck of high-dimensional grid volumes while capturing both local facies boundaries and macro-depositional geometry. It serves as the primary indicator of global architectural consistency and mode collapse during the training process. During training, the model saves 10% of its training data exclusively for evaluation using the MSSWD metric, ensuring these validation samples are never presented to the Discriminator network.

Beyond evaluating global structural realism, quantifying the spatial uncertainty and stochastic diversity across the generated ensemble is critical. To achieve this, voxel-wise Shannon’s Entropy (shannon1948mathematical) is utilized. Let N be the total number of realisations within an ensemble, and let $p_c(z, y, x)$ represent the empirical categorical probability of observing facies class $c \in \{1, \dots, K\}$ at a specific coordinate (z, y, x) , calculated as:

$$p_c(z, y, x) = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(\text{Realisation}_i(z, y, x) == c) \quad (3.3)$$

where $\mathbb{I}$ is the indicator function. The raw uncertainty at each coordinate is computed via Shannon's entropy field:

$$H_{\text{raw}}(z, y, x) = - \sum_{c=1}^K p_c(z, y, x) \log_2 p_c(z, y, x) \quad (3.4)$$

To ensure the metric is bound strictly between 0 and 1, facilitating direct cross-comparison between models with varying numbers of geological classes (e.g., 3-class versus 9-class setups), the field is normalized by the absolute theoretical upper bound $H_{\text{max}} = \log_2(K)$, which occurs under a uniform distribution of pure randomness:

$$H(z, y, x) = \frac{H_{\text{raw}}(z, y, x)}{\log_2(K)} \quad (3.5)$$

By calculating the normalized entropy at each specific spatial coordinate across multiple generated models, this metric serves as a direct validation of the model's stochastic diversity. High entropy values at a given pixel indicate that the generator produces a varied distribution of facies at that location, whereas low entropy exposes areas of rigid uniformity or localized mode collapse.

This voxel-wise evaluation visually maps regions of spatial conflict and agreement, which can be summarized across the entire volume domain to yield a global scalar score $\bar{H} = \frac{1}{V} \sum_{z,y,x} H(z, y, x)$. To evaluate how closely the spatial distribution of uncertainty matches the reference data, the coordinate-wise probability tensors are aggregated and compared using the **Jensen-Shannon Divergence (JSD)** (Lin1991divergence). The **JSD** measures the coordinate-wise distributional disagreement between the reference baseline and the **GAN** models (The full mathematical derivations and multi-scale details for these evaluation metrics are expanded in Appendix D.2).

Data conditioning

While JS divergence defines the overall ensemble spread, after optimization and inversion, each individual **GAN** model within that ensemble must still strictly honor the physical reality at the wellbore. This physical reality at the wellbore acts as a strict spatial constraint, the localized mismatch of which is calculated using the Macro F1-score. The Macro F1-score calculates the unweighted average of the harmonic mean of precision and recall across all C individual facies classes:

$$F1_{\text{macro}} = \frac{1}{C} \sum_{c=1}^C \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}$$

To evaluate this localized mismatch, the precision and recall for each specific facies class (c) are first defined as:

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c}$$

Here, a true positive (TP_c) represents the number of correctly predicted voxels for the target facies, a false positive (FP_c) represents voxels incorrectly assigned to the target facies, and a false negative (FN_c) represents actual target facies voxels that the model failed to predict.

In addition to the class-specific F1-scores, the overall categorical match accuracy is computed to provide a high-level metric of the model's spatial reliability. Accuracy evaluates the total proportion of correctly predicted voxels across all classes relative to the total number of evaluated wellbore voxels, formally defined as the sum of all true positives divided by the total volume of conditioning points (**Google2025AccuracyPrecision**).

4 Results

The presentation of the results will follow the order of the workflow: First of all the curated datasets will be displayed with the addition of statistical parameters that can be used to evaluate the difference between the samples in the datasets and the synthetic GAN samples. Secondly, the post training results summarize the results of the hyperparameter tuning, the Convolutional Generative Adversarial Network (cGAN) unconstrained performance to grouped facies classes as well as the cGAN unconstrained performance when presented with all facies classes. Finally, the results of the optimization and inversion results are presented, after the model was constrained to the Delft Geothermal Reservoir well data.

4.1. Datasets

In total, 20,000 samples were generated for each of the three datasets as defined in table 3.1. The generated datasets are visually summarised in figure 4.1: a 2D Multi-Dimensional Scaling (MDS) plot. MDS is a technique used to simplify complex, high-dimensional data into a clear two-dimensional map. It looks at the differences between the 3D grid samples and plots them so that similar realisations sit close together, while vastly different ones are placed far apart **kruskal1964multidimensional; rongier2025atowards**. This plot clearly shows a grouping effect per setting, indicating the samples of different settings do in fact display varying higher-dimensional differences. This can be visually confirmed by the two random 3D facies plots per setting, showing that a lower sand body extension as input parameter yields samples with thinner sandbodies and more erratic meandering behaviour in samples of this size. The right-most plots show the pixel-wise entropy computed over an ensemble of 1000 samples per setting, setting a baseline for the entropy readings we look to achieve from a diverse synthetic dataset produced by the trained GAN model. This threshold configuration can be further evaluated by analysing table 4.1, which provides an exhaustive overview of the global facies distributions across the ensemble. As can be observed in this overview, the structural facies, exhibit a severe statistical imbalance

Facies	Setting 1 (%)	Setting 2 (%)	Setting 3 (%)
1: Channel Lag	1.6	1.8	1.8
2: Point Bar	67.5	66.3	65.4
3: Sand Plug	1.8	1.6	1.4
4: Crevasse Splay Core	0.0	0.0	0.0
5: Crevasse Channel	0.1	0.1	0.1
6: Crevasse Splay Delta	0.0	0.0	0.0
7: Levee	5.1	6.5	7.7
8: Overbank	10.2	11.4	11.9
9: Mud Plug	13.6	12.4	11.5

Table 4.1: Global categorical facies distribution profiles compared across the three synthetic training datasets for all available classes.

ance throughout the generated data volume. This imbalance stems from dominant facies occupying over 60% of the total grid volume, while specific minority facies are heavily suppressed, highlighting a limitation of the FLUMY NEXUS parameters. Specifically, the volumetric presence of facies 4, 5, and 6 is negligible, yielding an exceptionally low occurrence rate ranging between 0.0% and 0.1%. Such low occurrence rates present a optimization challenge during adversarial training. Because standard GAN objective functions treat all voxels with uniform weight, the discriminator may be biased to evaluating sample validity almost entirely on the dominant Class 1. As a result, the generator can get trapped in a suboptimal training loop that refines large-scale features while completely disregarding the struc-

tural nuances of minority facies. This statistical skewness must be strictly accounted for during model evaluation, particularly regarding the localized well mismatch metrics.

When transitioning from the detailed 9-class case to the upscaled 3-class case, the secondary class naturally increases to make up about 5% of the total volume. Quite by chance, these grouped facies proportions closely match the actual empirical proportions interpreted from the real cored and logged sections of the DEL-GT-01 and DEL-GT-02-S2 wells (see Figure A.1). Therefore, while the original 9-class prior struggles with severe generation limitations under default settings, this upscaled version accidentally aligns the model’s facies distributions perfectly with the target reservoir.

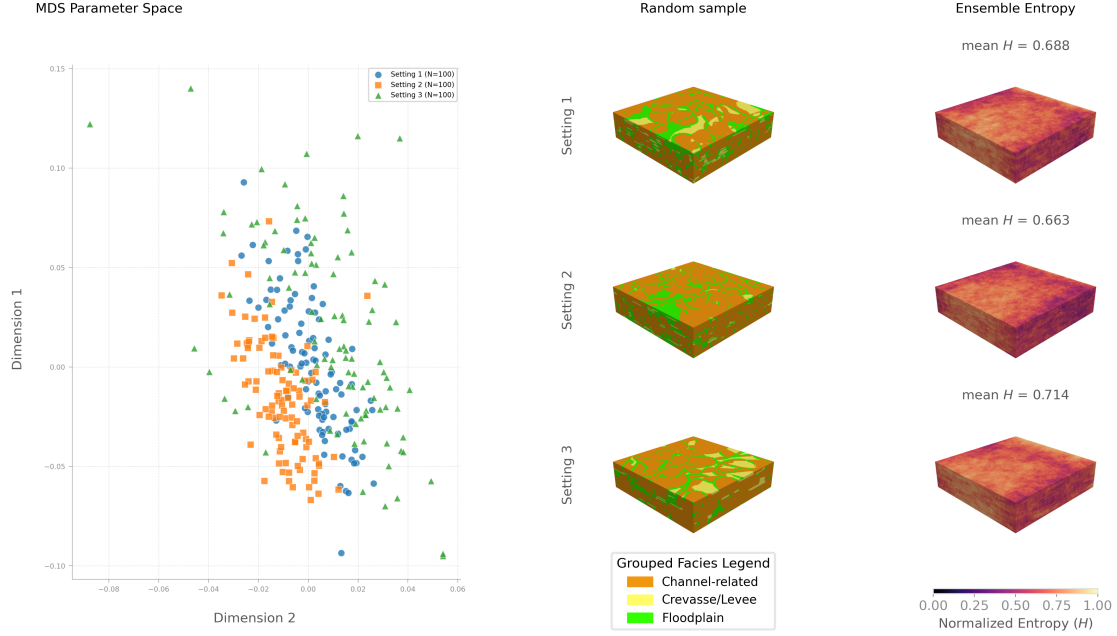


Figure 4.1: MDS plot with two 3D facies samples displayed and a 3D pixel-wise entropy per setting.

4.2. Post Training Generative Results

All the models were trained on 2 NVIDIA-A100 GPU’s with each 80GB of unified memory. When trained for 25 epochs on 20000 samples, the execution time took approximately 3-4 hours.

4.2.1. Hyperparameter Tuning

Figure 4.2 and table 4.2 provide an overview of the hyperparameter tuning study. All trained model reserved 10% of their training data exclusively for validation: computing the [Sliced Wasserstein Distance \(SWD\)](#) metric for both training and validation datasets. Figure 4.2 includes tracking of the SWD metric across 25 epochs, a randomly generated sample by the trained model and a 3D entropy plot of an ensemble of 1000 [GAN](#) generated samples. Complementary, the [JSD](#) and [MS-SWD](#) which both compare the ensemble of 1000 generated samples per model to a test set of 1000 samples is provided in table 4.2.

Based on the metrics from table 4.2 and entropy plots from figure 4.2, the best model was achieved in model 5 with a [JSD](#) of 0.0155, [MS-SWD](#) of 0.07986 and with the best voxel-wise entropy value out of all the models (0.79), supported by the 3D entropy plot closely resembling the 3D entropy plot of the setting 2 test dataset. The [MS-SWD](#) metric showed all samples decreasing in sliced wasserstein’s distance, indicating that the generators are quickly learning to produce samples that have reduced SWD compared to the validation dataset. An interesting observation with regards to the SWD evolution over 25 epochs can be made: All plots show a stabilizing SWD after 2000 training iterations. From a SWD metric standpoint, this indicates that the model has stopped improving its samples compared to the validation dataset.

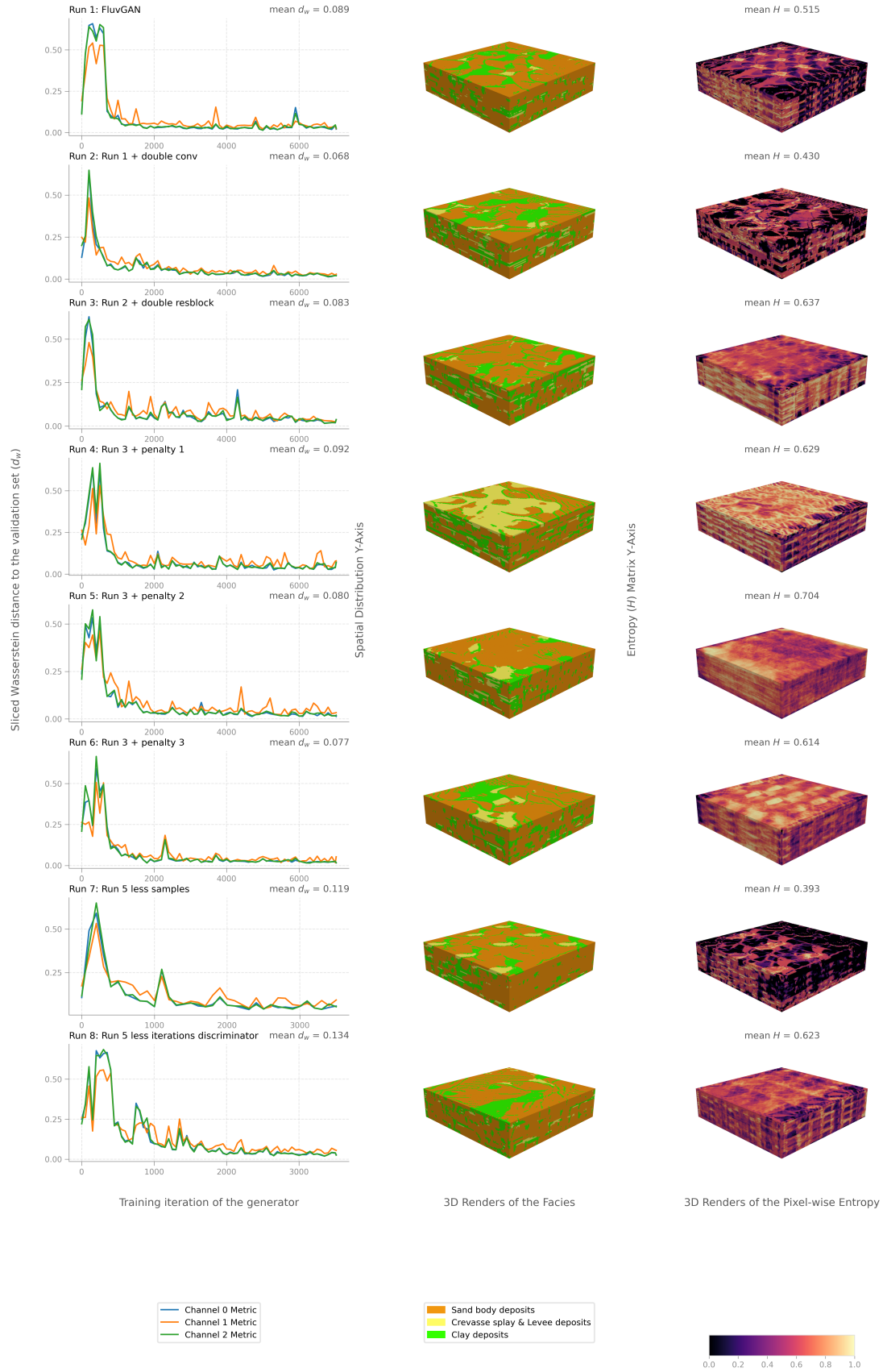


Figure 4.2: Ablation study starting from FluvGAN and progressively adding/adjusting parameters to observe their effect on training, geological plausibility and sample diversity. *model 1*: Standard FluvGAN architecture from (rongier2025atowards). *model 2*: Addition of double convolutional layers. *model 3*: Addition of double residual blocks (i.e. skip connections) on top of double convolutions. *model 4*: Addition of a discriminator penalty every 32 iterations. *model 5*: CHanged discriminator penalty to every 16 iterations. *model 6*: Changed discriminator penalty to every iteration. *model 7*: model 5 hyperparameters but decreased to 10000 training samples. *model 8*: model 5 hyperparameters but changed number of iteration of the discriminator to 2.

model	Epochs	disc penalty	double conv	double resblock	JSD ↓	MS-SWD ↓	mean H_{norm} ↑
1	25	no	off	off	0.0718 ± 0.0157	0.08865	0.515
2	25	no	on	off	0.1161 ± 0.0243	0.06842	0.430
3	25	no	on	on	0.0458 ± 0.0076	0.08271	0.637
4	25	32	on	on	0.0612 ± 0.0314	0.09224	0.629
5	25	16	on	on	0.0155 ± 0.0067	0.07986	0.704
6	25	1	on	on	0.0202 ± 0.0038	0.07690	0.614
7	25	16	on	on	0.1190 ± 0.0324	0.11924	0.393
8	25	16	on	on	0.0278 ± 0.0045	0.13389	0.623

Table 4.2: Overview of hyperparameters and evaluation metrics of each model. model 5, based on architecture 4 by (rongier2025atowards) yields the best JSD and second to best MS-SWD value.

Moreover, the base FluvGAN architecture lead to significant mode collapse looking at figure 4.2. Only after increasing the model depth (double convolutions & double residual blocks) did the model train properly without mode collapse. Moreover, when varying the discriminator R1 Regularization, applying the penalty every 16 iterations yielded the best result. All other option (no penalty, every 32 iterations and every iteration) all resulted in mode collapse. Finally, model 7 and 8 investigated whether decreasing the number of training samples from 20000 to 10000 and only updating the discriminator every 2nd iteration, yielded any improved results. Where the results from model 8 yielded similar results to model 6, the metrics from model 8 yielded a much higher mean JSD of 0.1190 ± 0.0324 . Additionally, looking at the entropy plot of model 8 in figure 4.2, The plot shows a significantly higher percentage of regions with low entropy values, indicating that most samples produced by the generator after training have similar pixel-wise values.

4.2.2. cGAN Unconstrained Performance

To evaluate if features continue to refine during late-stage training, Model 5.1 extended the baseline Model 5 configuration to 100 epochs. Figures 4.3 and E.2 show the mid-slices along the X, Y, and Z axes for three generated samples compared to the test dataset. Both models successfully generate meandering sand bodies with central mudplugs (B.1), alongside adjacent floodplains and levees. Additionally, the X- and Y-axis plots display “V” shaped mudplug infills and elongated clay deposits. However, the slices also reveal structural artifacts, including closed loops ①, previously noted by sun2023geological and new isolated random voxels ②. Since 2D visual inspection cannot capture the 3D spatial continuity of these features, quantitative 3D metrics are necessary to properly evaluate model performance.

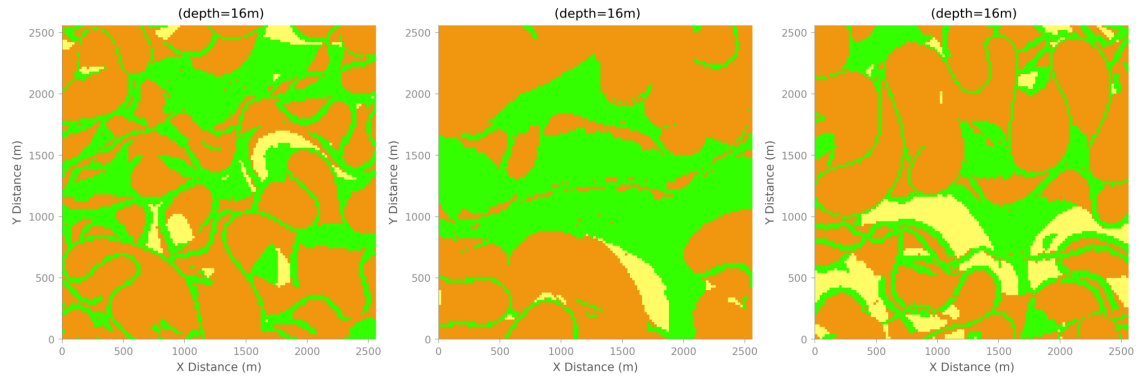
table 4.3, provides an overview of the main evaluation metrics used on models 5 and 5.1. Both models score almost equal on all metrics: the JSD metric improved slightly with more training epochs from 0.0146 to 0.0132 whereas the mean voxel-wise entropy decreased from 0.704 to 0.684. Both models show a clear mismatch in unique patterns present in the ensembles with a $\pm 30\%$ increase over the baseline. Furthermore, Figure C.2 shows no significant structural shift in the MS-SWD multidimensional scaling. The clustered distributions indicate that the extra training epochs fail to improve the model’s overlap with the test dataset, making the two models hardly distinguishable under this metric.

name	epochs	JSD ↑	MS-SWD ↓	mean H_{norm}	n unique patterns
setting 2	-	-	-	0.692	2,949,220
model 5	25	0.0146	0.08	0.704	3,846,513
model 5	100	0.0132	add	0.684	3,727,836

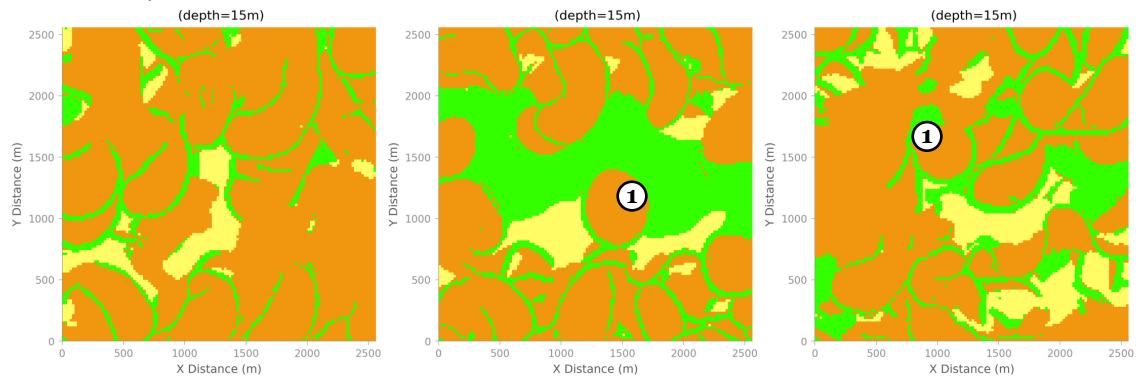
Table 4.3: Overview of hyperparameters and evaluation metrics of each model ($N = 100$ realizations). Model 5, based on architecture 4 by (rongier2025atowards), yields the best JSD and second to best MS-SWD value.

The connectivity and pattern analysis metrics were computed to evaluate the facies conductivities compared to the baseline, as well as assessing the number of unique patterns. Figure 4.4 showcases the facies connectivity distributions for both models. While the general overlap in connected component

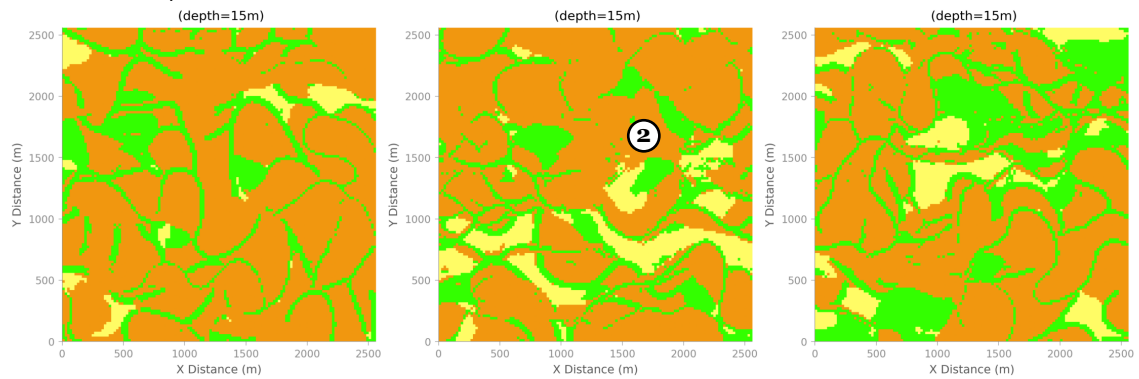
FLUMY setting 2 samples



Model 5 samples



Model 5.1 samples



Sand body deposits
 Crevasse splay & Levee deposits
 Clay deposits

Figure 4.3: 2D slices taken from the Z axis at depth 16m of samples from 1) the *FLUMY* test dataset, 2) Model 5 (trained for 25 epochs) and 3) model 5.1 (trained for 100 epochs). ① denotes areas of closed loop structures whereas ② denotes areas with single isolated voxels.

(blob) sizes relative to the test dataset indicates that the models successfully capture facies groupings, systematic discrepancies remain. Both configurations overestimate the frequency of isolated single voxels for the Crevasse Splay & Levee deposits, consequently underestimating the occurrence of larger clusters containing 10 or more voxels. A similar trend is observed in the Clay deposits generated by Model 5.1, which exhibits a corresponding overestimation of isolated single voxels. This phenomenon can be the explanation for the mismatch in number of unique patterns present in the ensembles, where the isolated voxels result in a substantial number of additional unique patterns. Although a minor difference, model 5.1 actually performs slightly better than model 5 on the larger blob sizes (range from 10^1 - 10^3) in both the crevasse splay & levee deposits as well as the clay deposits, suggesting that the increased number of epochs dedicated to training helped the model in better predicting facies continuity.

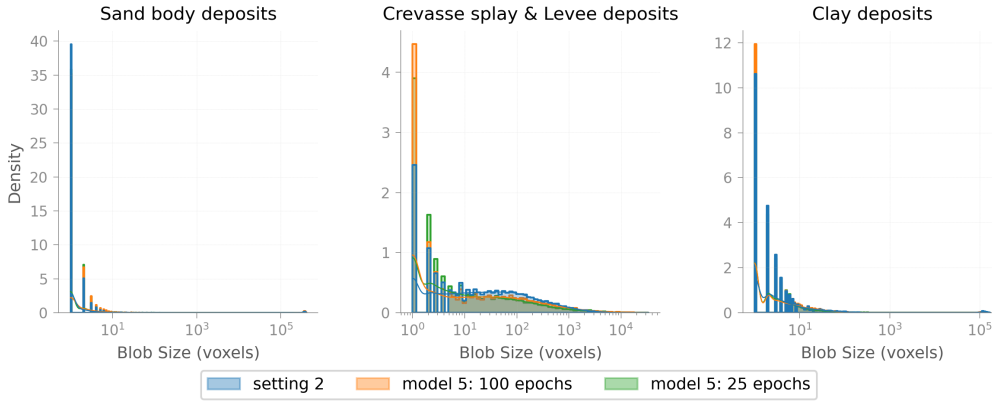


Figure 4.4: Connectivity analysis plot of the test dataset, illustrating the ability to recreate blob sizes of model 5 for 25 and 100 training epochs.

Finally, the JSD normalised voxel-wise entropy was evaluated across all three axes (figures 4.5 and E.1). In the baseline dataset (setting 2), entropy fluctuates significantly along the X and Y axes, yet remains highly stable along the Z-axis. When evaluating the generative models, Model 5 successfully mirrors this Z-axis baseline consistency. In contrast, Model 5.1 exhibits a systematic decline in entropy along the Z-axis. This indicates that prolonged training (Model 5.1) restricts the generative variance, suggesting that extended epochs induce a partial mode collapse rather than an improved representation of spatial diversity for this case study.



Figure 4.5: Comparison of the JSD normalized voxel-wise entropy across the Z, Y, and X axes. The decline in Z-axis entropy for Model 5.1 highlights a loss in stochastic diversity.

The voxel-wise entropy across the horizontal axes also reveals a trend in the original training datasets. Across both the X and Y axes, the normalised entropy varies significantly, exposing a heavy spatial bias within the FLUMY simulations that leads to a non-stationary geological prior. Along the Y-axis, the baseline displays a inverted U-shape; entropy is suppressed at the grid boundaries ($Y = 0$ and $Y = 128$)

and peaks in the center. This indicates a clear channel-belt constraint, where boundary conditions force the simulated river into a central corridor, leaving the edges highly predictable and lacking architectural variety. Similarly, the X-axis exhibits a distinct U-shaped curve with a sharp drop in entropy around slices 20 to 30. This signature points to a fixed sediment inlet or nodal avulsion point, where the river channel is physically restricted to a narrow path before expanding downstream. Consequently, while the vertical depositional energy remains perfectly stable across the Z-axis, the horizontal baseline prior is heavily biased by the deterministic rules of the process-based simulator.

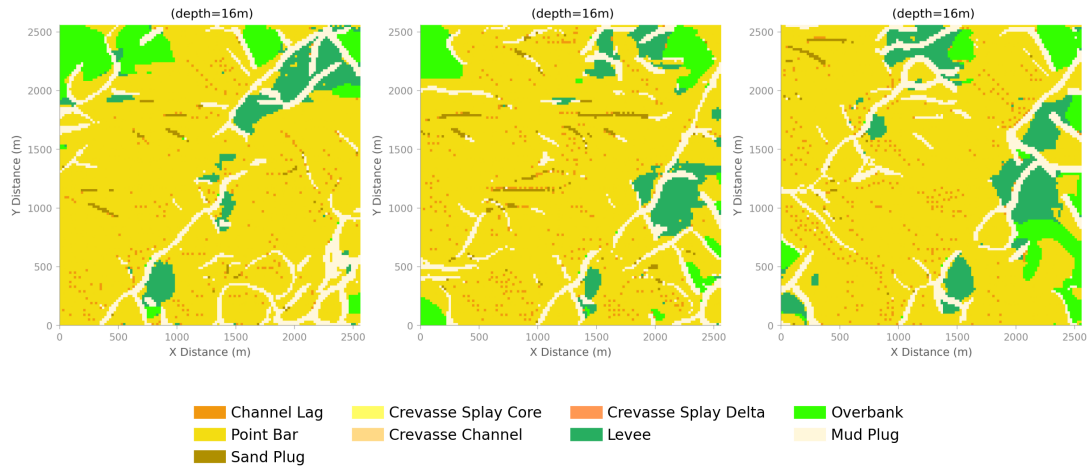
4.2.3. High-Fidelity and Multi-Prior Models

To test the limits of the generative framework, two additional scenarios were evaluated: a high-fidelity model and a multi-prior model.

The high-fidelity model was trained on the same baseline dataset as before (Setting 2), but it retained all nine individual facies classes instead of grouping them into three FAs. Because the nine original classes are highly unbalanced, this scenario tests how well the generator handles severe class imbalances.

In contrast, the multi-prior model kept the grouped three-class configuration but was trained on a combined dataset containing 10,000 samples from each of the three different geological settings (Settings 1, 2, and 3). This setup evaluates whether the Model 5 hyperparameters can successfully handle a mixture of different geological rules. Both models were trained for 25 epochs, and the results are summarized in Table 4.4.

Model 5 - high-fidelity



Model 5 - all settings

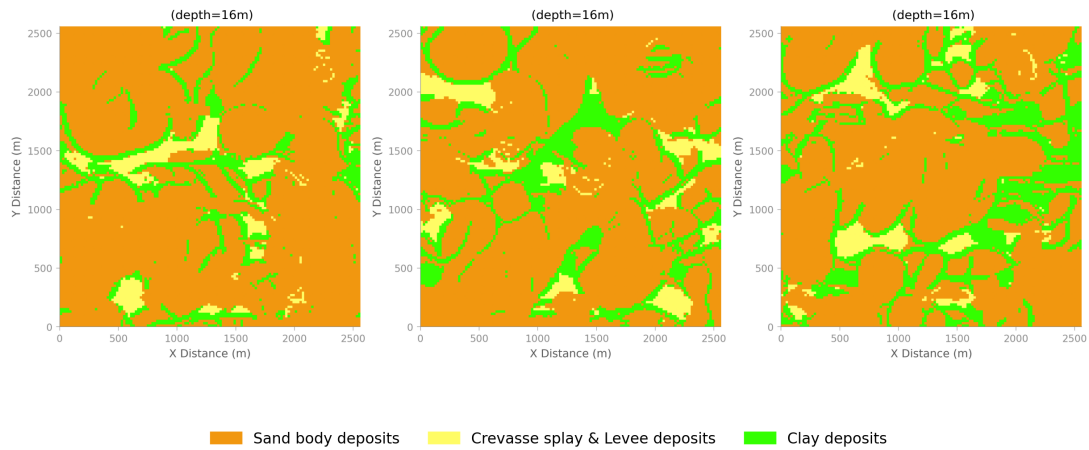


Figure 4.6: 2D slices taken from the Z axis at depth 16m comparing the high-fidelity model (9 classes, Setting 2) and the multi-prior model (3 classes, combined settings).

Model Configuration	Epochs	JSD ↓	MS-SWD ↓	Mean H_{norm} ↑	n Unique Patterns
Setting 2 (grouped)	-	-	-	0.692	2,949,220
Setting 2 (non-grouped)	-	-	-	0.695	8,864,195
High-Fidelity (9-Class, Setting 2)	25	0.1102		0.351	6,017,855
Multi-Prior (Grouped, All Settings)	25	0.0168		0.661	4,512,631

Table 4.4: Overview of evaluation metrics comparing the high-fidelity and multi-prior configurations against the baseline implementations ($N = 100$ realizations).

The high-fidelity model is able to generate samples with all nine facies distribution correctly represented in them (figure ??). From a visual perspective we can however see a high number of single isolated voxels belonging to the channel lag facies, as well as a mud plug and sand plug facies with bad connectivity and structures that do not fully resemble geologic realistic structures. The pixel-wise entropy plot (figure 4.7) shows a significant amount of rigid structure, indicating that the model lacks spatial entropy and cannot capture the full complexity of the training data. This is confirmed by a mean pixel-wise entropy value of 0.351, significantly lower than the baseline mean normalized entropy achieved.

Conversely, the multi-prior model suffers from heavy mode collapse. When presenting the Model 5 hyperparameters with the combined dataset of all three settings, the training process destabilizes. As shown in Figure 4.6, the generator stops producing diverse geological variations and instead outputs nearly identical, repetitive patterns that fail to represent the unique features of the different input settings.

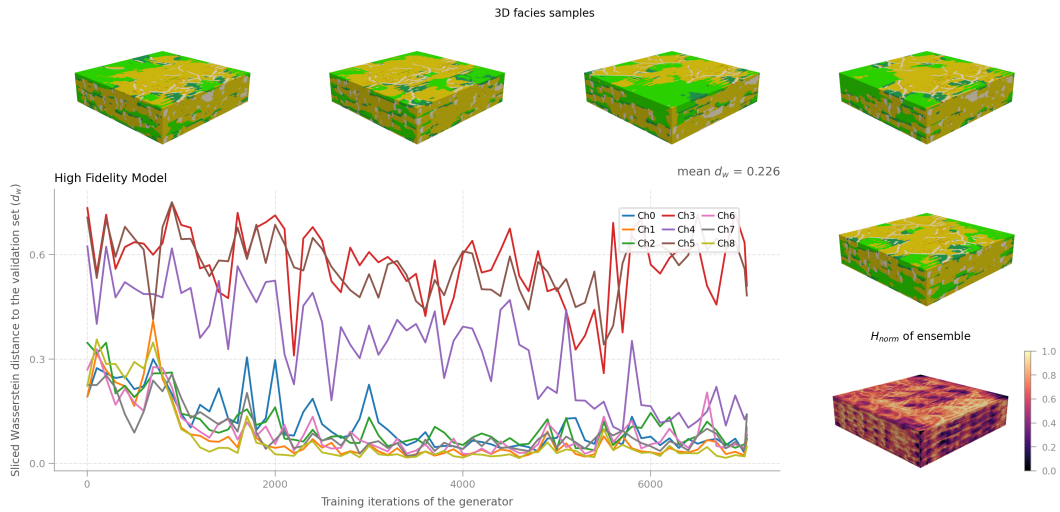


Figure 4.7: Overview of the high-fidelity model metrics and training evolution over 25 epochs.

The SWD evolution for the high-fidelity model shows that performance varies significantly by channel. For channels 0–2 and 6–8, the SWD metric decreases immediately from the start of training, showing that the model learns these structures well. However, for channels 3 and 5, the metric shows no improvement, meaning the model fails to represent these specific minority classes. Channel 4 decreases steadily over the full 25 epochs, suggesting it might improve further with more training. Meanwhile, channels 0, 2, and 6 maintain low initial values with occasional minor increases.

These outcomes demonstrate that while the Model 5 configuration was highly effective for the 3-class, single-setting case, directly expanding it to the 9-class and multi-prior scenarios does not achieve stable or diverse GAN convergence. It should be noted that no separate hyperparameter tuning or ablation study was performed specifically for these two complex scenarios. Consequently, while Model 5 pro-

vides a strong baseline, these findings suggest that higher-dimensional or mixed-prior datasets require dedicated hyperparameter optimization to prevent destabilization and mode collapse.

4.3. Post-Inversion Results

During inversion, the tuned model was conditioned to the well data presented in Section 2.1. The evaluation metrics were computed exclusively on the first 32 metres of well data from DEL-GT-01 and DEL-GT-02-S2, yielding a total of 64 spatial conditioning points for the Macro F1-score evaluation. This specific interval provides a good testing ground because it captures a highly contrasting geological setup between the two well locations. Within this 32-metre window, the wells display completely different facies distributions: DEL-GT-01 contains almost no Overbank Mudstone, whereas DEL-GT-02-S2 contains virtually no Levee deposits. Choosing this asymmetric dataset serves as a rigorous proof-of-concept. It allows us to assess whether the inversion process can successfully adapt to distinct local constraints across different areas of the grid, rather than just learning a single, uniform rule. During the initial Latent Space Optimization phase, 100 latent vectors were optimized to fit the well data. These outputs are referred to as “realisations”, as they represent models explicitly conditioned to real-world subsurface observations.

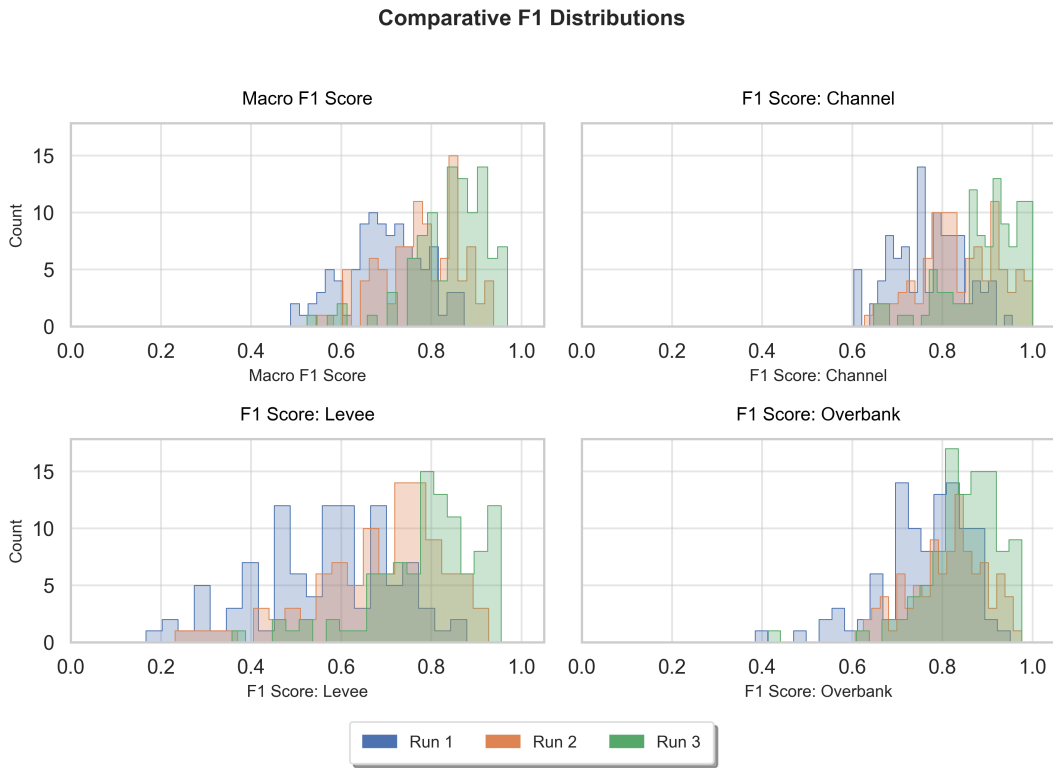
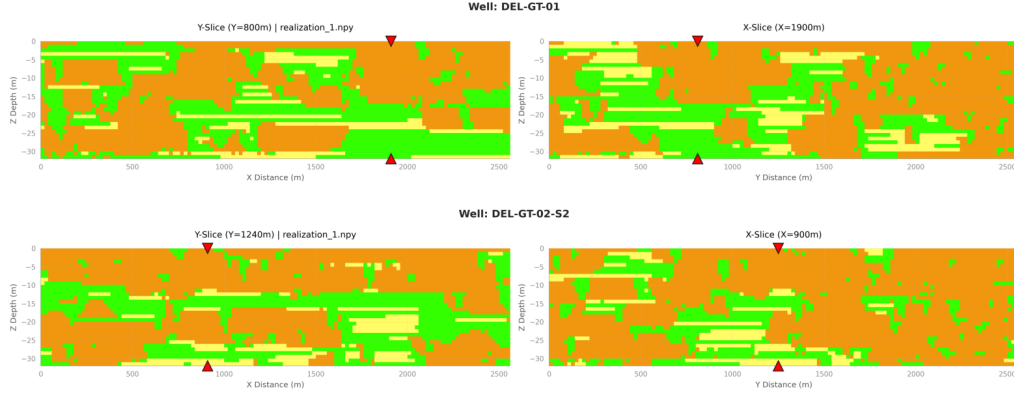


Figure 4.8: Macro F1 and single channel F1 score distributions for three optimization models with varying threshold values. model 1: threshold = 1, model 2: threshold = 2 & model 3: threshold = 5.

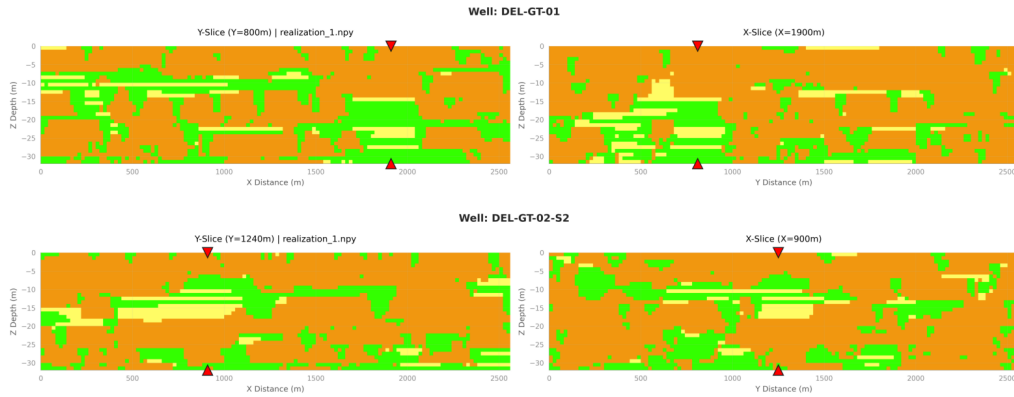
As illustrated in Figure ??, the performance of realisations achieving a tight fit with the conditioning points is highly dependent on the optimization threshold (Th). The threshold (Th) defines a specific spatial radius around the wellbore, measured in voxels, within which the generative model is penalized for mismatching the hard conditioning data. Voxels located beyond this distance threshold are assigned a weight of zero in the context loss function, ensuring that the inversion process enforces localized geological alignment without over-constraining the broader, unobserved reservoir architecture (See figure D.1). Table 4.5 details the Macro F1-scores, per-class F1 performance, and mean spatial entropy metrics achieved across these varying threshold configurations.

The (macro) F1-score distributions of realisations exhibit a positive response to the threshold parameter. Moving from a tightly constrained threshold of 1 (model 1) to a wider threshold of 5 (model 3)

Run 1: Threshold 1 (X- and Y- Axis Sections)



Run 1: Threshold 2 (X- and Y- Axis Sections)



Run 1: Threshold 5 (X- and Y- Axis Sections)

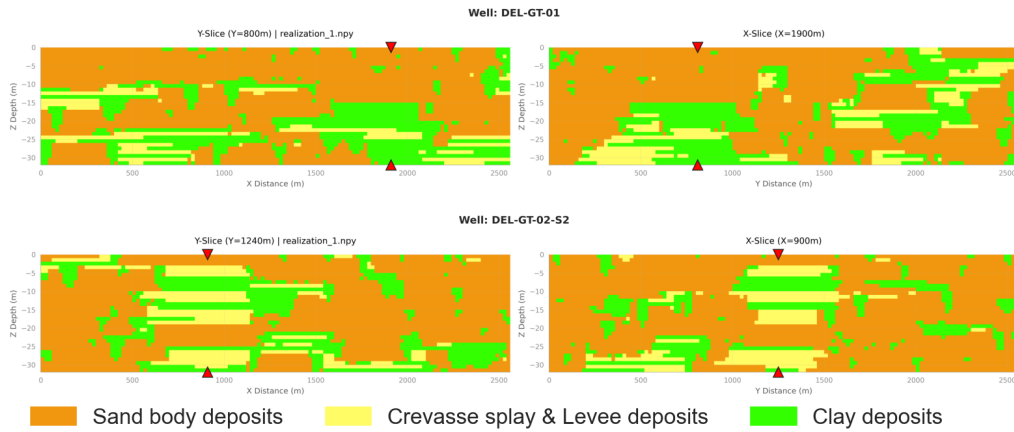
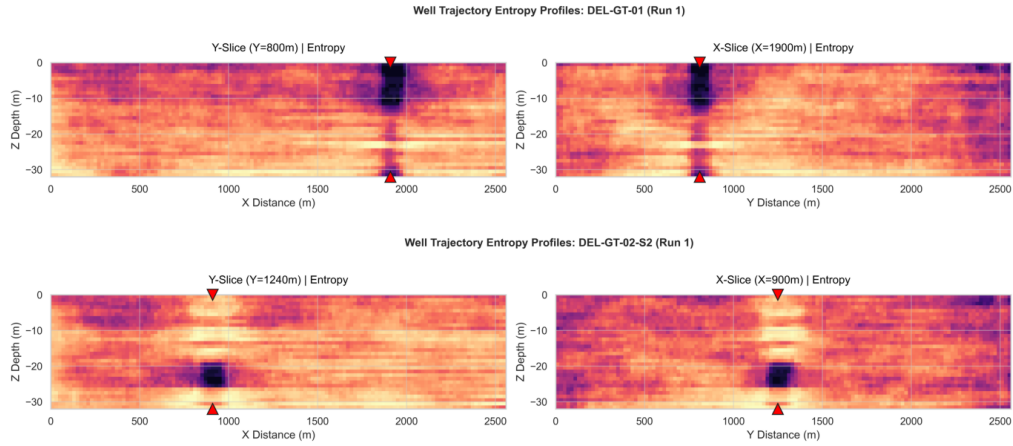
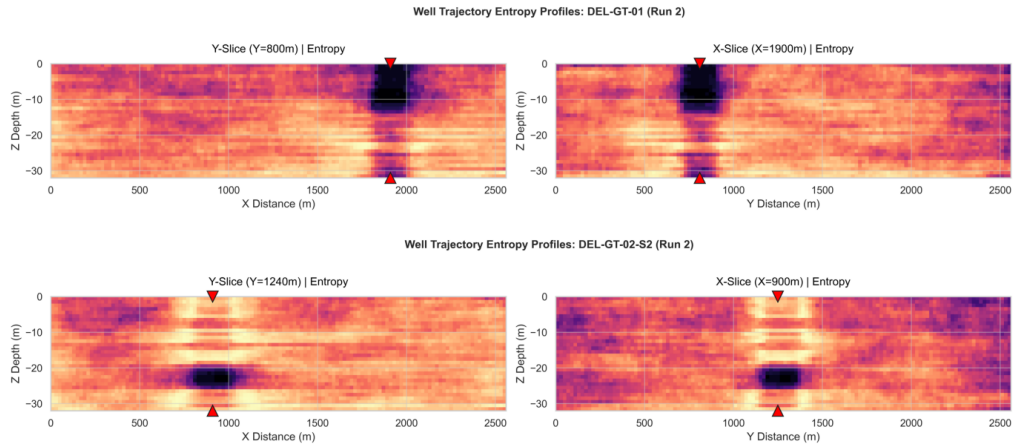


Figure 4.9: 2D slices taken over the X- and Y- axes at the well trajectory coordinates realisations generated after latent optimization. The regions nearby the well trajectory show continuity of geologic features when found in the wells.

Run 1: Threshold 1 (Entropy: X- and Y- Axis Sections)



Run 1: Threshold 2 (Entropy: X- and Y- Axis Sections)



Run 1: Threshold 5 (Entropy: X- and Y- Axis Sections)

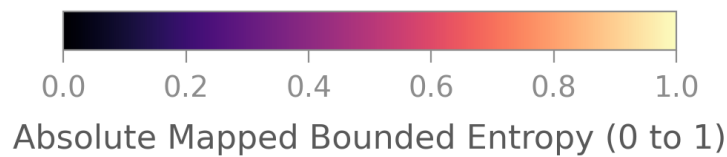
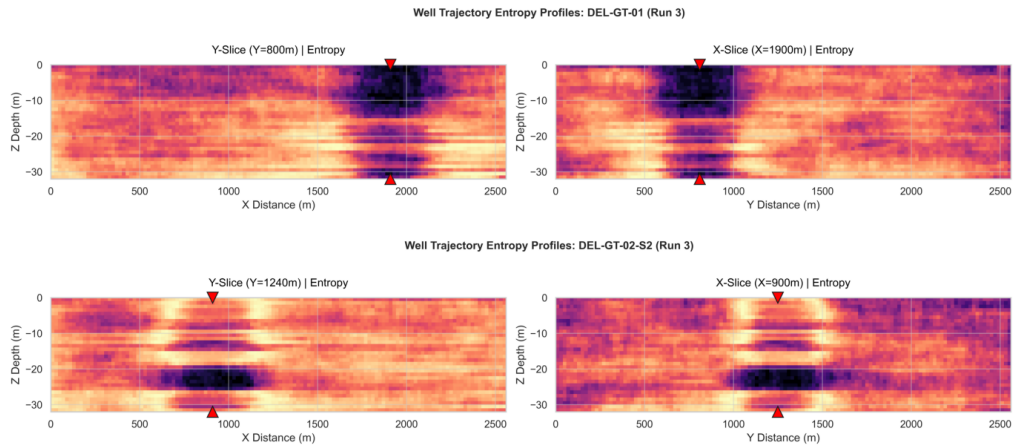


Figure 4.10: 2D entropy slices taken over the X- and Y- axes at the well trajectory coordinates realisations generated after latent optimization. The regions nearby the well trajectory show an increased low entropy zone as the threshold increases. Moreover, the entropy decreases is mainly caused by the presence of the clay deposit, which often indicates the presence of a floodplain.

model	Th	mean Macro F1 $\uparrow$	mean $F1_{C1}$ $\uparrow$	mean $F1_{C2}$ $\uparrow$	mean $F1_{C3}$ $\uparrow$	mean H_{norm} $\uparrow$
model 1 (lso)	1	0.70	0.77	0.56	0.76	0.686
model 2 (lso)	2	0.78	0.84	0.68	0.81	0.685
model 3 (lso)	5	0.84	0.90	0.79	0.85	0.663
model 1(pv)	1	0.89	0.8989	0.87	0.89	0.7137

Table 4.5: Optimization metrics evaluated across different model configurations where the lso indicates post latent optimization and (pv) post pivotal tuning results. The column ‘Th’ indicates the specific threshold value defining the spatial region affected by latent optimization. Per-class F1-scores ($F1_{C1}$, $F1_{C2}$, $F1_{C3}$) indicate conditioning accuracy across the individual facies categories at the well locations, accompanied by the global mean normalized spatial entropy (H_{norm}) computed over the generated ensemble.

yields a substantial increase across all metrics, with the Macro F1-score rising from 0.70 to 0.84. This upward shift is reflected consistently across all three facies classes. The dominant class ($C1$) scales from an F1-score of 0.77 up to 0.90, while the minority class ($C2$), which typically represents the optimization bottleneck due to severe class imbalance, sees its F1-score jump from 0.56 to 0.79. This systemic improvement suggests that expanding the spatial radius of the context loss mask provides the latent space optimization with the necessary information to allow the model to find valid spatial arrangements that honor the hard conditioning points.

To evaluate the conditioning performance down the vertical profiles, Figure 4.11 displays the depth-by-depth ensemble prediction probabilities for each individual facies alongside the comprehensive categorical match accuracy. The vertical profiles demonstrate a clear correlation between the framework’s generative confidence and the global class distributions of the prior training dataset.

For depth intervals corresponding to the dominant sand body deposits (FA1), the ensemble channel probability consistently stays pinned close to 1.0, translating to an almost perfect overall match accuracy across the realization ensemble. Conversely, the highly underrepresented crevasse splay and levee deposits (FA2) expose a significant optimization bottleneck; the ensemble prediction probability for FA2 peaks at only 0.7, with some runs dropping as low as 0.3 even within intervals where the facies is explicitly logged. The clay deposits (FA3) exhibit intermediate performance characteristics, demonstrating reasonable structural replication and stability across the vertical column.

This stark performance disparity directly mirrors the underlying class imbalance present within the foundational FLUMY datasets, where FA1 is the most abundant facies followed by FA3, while FA2 remains severely suppressed. The dominant volumetric occupancy of FA1 allows the generator to map its structural boundaries with high confidence, whereas the sparse representation of FA2 prevents the latent space optimization from establishing highly confident, deterministic boundaries along the wellbore paths.

Furthermore, analyzing the relationship between conditioning accuracy and ensemble variance highlights a spatial trade-off during the latent space optimization phase. The main takeaway is that as the global F1-score increases via a wider threshold, the normalized spatial entropy (H_{norm}) systematically decreases. Geologically and mathematically, this behaviour occurs because the area under direct deterministic constraint increases radially from the wellbore with higher threshold values. By expanding this optimization zone, a larger volume of the reservoir is forced to adhere strictly to the static well data across the entire ensemble. Consequently, the area characterized by low entropy expands radially outward from the well trajectories, naturally diminishing the overall stochastic diversity and structural variance of the generated realisations in that region (see Appendix ??).

While expanding the context loss threshold improves conditioning accuracy, the associated penalty to stochastic diversity highlights the critical necessity of the second phase in the inversion workflow: pivotal tuning. Because latent space optimization often struggles to find a perfect voxel-by-voxel match in a frozen latent space—especially for heavily imbalanced datasets—pivotal tuning temporarily unfreezes

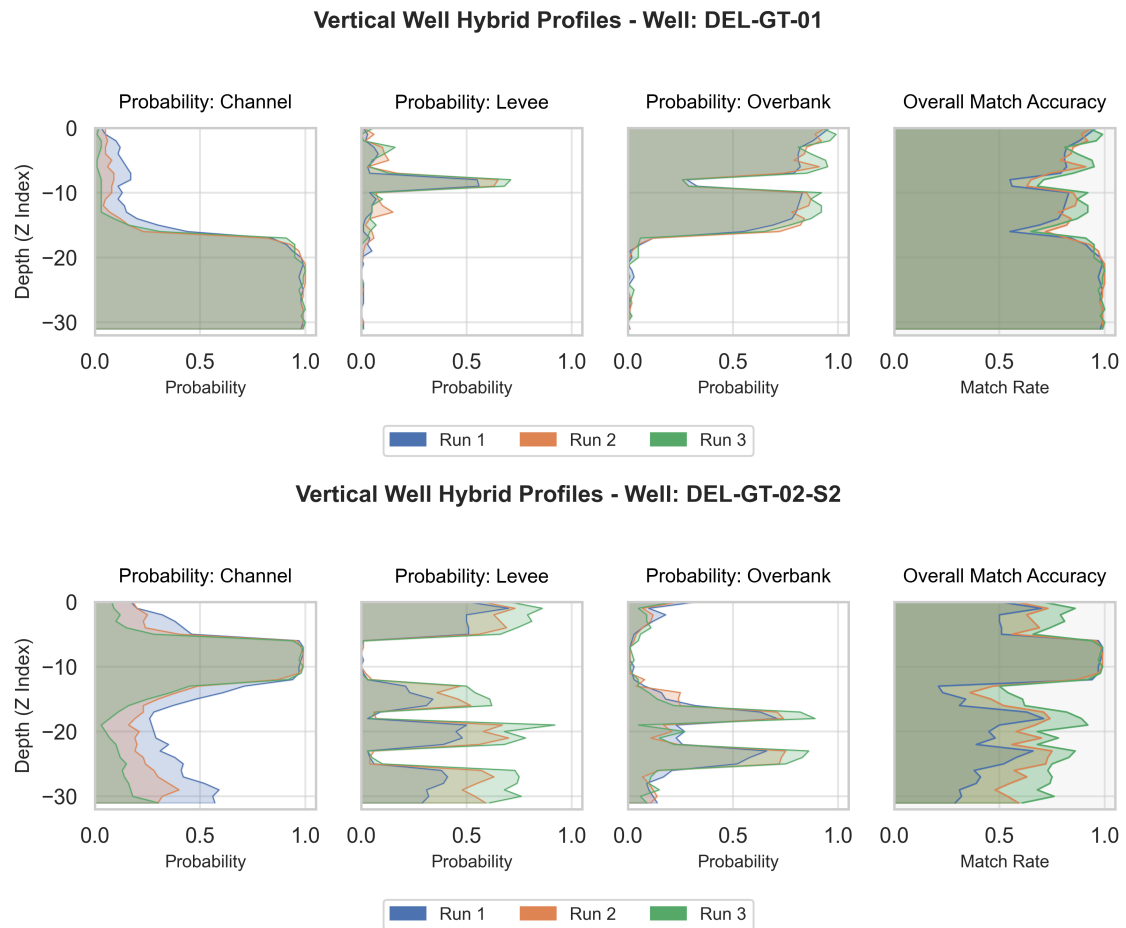


Figure 4.11: Vertical depth profiles for wells DEL-GT-01 and DEL-GT-02-S2 showing the individual ensemble prediction probabilities per facies (left three panels) and the aggregate categorical match accuracy (rightmost panel) evaluated across three optimization models (Model 1: threshold = 1, Model 2: threshold = 2, and Model 3: threshold = 5).

the generator’s internal weights to forcefully align the output with the well data.

As detailed in Table 4.5, applying pivotal tuning to the tightly constrained baseline (model 1, $Th = 1$) yields **exceptional** improvements. The mean Macro F1-score surges from 0.70 under latent space optimization alone to 0.89 post-pivotal tuning (model 1(pv)). Most notably, this phase effectively resolves the optimization bottleneck associated with the severely suppressed minority class ($C2$). The F1-score for $C2$ experiences a dramatic jump from 0.56 to 0.87, easily outperforming even the widest latent optimization threshold ($Th = 5$, which only achieved 0.79).

This massive localized accuracy increase bypasses the spatial variance penalty observed when simply widening the [Latent space optimizations and pivotal tuning \(LSO\)](#) threshold. Instead of dropping, the mean normalized spatial entropy (H_{norm}) after pivotal tuning actually increases to 0.7137. This indicates that by making microscopic, localized adjustments to the generator’s weights, pivotal tuning successfully forces deterministic alignment exactly at the wellbore without over-constraining the surrounding grid. As a result, the model achieves near-perfect conditioning to sparse real-world data while safely preserving the stochastic diversity of the broader, unobserved reservoir architecture.

5 Discussion & Recommendations

The application of **cGANs (GANs)** for geomodelling is a logical design choice given the substantial research and code repositories available (**sun2023geological**; **song2023gansim**). Previous works demonstrated that generating realistic 2D slices with a **GAN** is viable, while recent studies extended this to 3D volumes to condition model realizations to well data (**bhavsar2024stable**; **rongier2025atowards**). However, these existing studies relied exclusively on synthetic training and well data. To address this limitation, this work evaluates the workflow using a real-world case study of the Delft Geothermal Reservoir

In this final chapter, the applied workflow is discussed: assessing the strengths as well as highlighting the weaknesses and consequently, providing the main recommendations in terms of model design. Finally, the recommendations regarding implementing the workflow for real-world purposes is provided.

5.1. Interpretations of key results

The proposed **FluvGAN** model has shown to be able to recreate plausible geological structures in a 3D domain, **while retaining sample diversity**, given a good balance has been found w.r.t. model hyperparameters. The presence of meandering and erosional features in the generated samples indicates that the 3D convolutions successfully capture dependencies across the entire volume, effectively learning both horizontal (XY) and vertical (Z) geological relationships. Additionally, when taking an ensemble of samples generated by the **FluvGAN** model, the facies proportions are replicated almost perfectly. During the inversion process, the combination of latent space optimization and pivotal tuning proved to be successful in achieving a mean **accuracy score** ranging from 0.78 to 0.89 on the generated samples on the hard well data. More importantly, the region affected by the inversion process still shows good diversity in the generated geological structures with low influence on the overall ensemble entropy. Inspecting the generated samples in figure 5.1 shows that the model is seemingly struggling with reproducing the training dataset with similar 'smoothness': areas with single isolated voxels of the clay deposits as well as continuous mudplugs at times being broken up by some voxels classified as sand body deposits, after which the mudplugs continue again. This suggests that the model is having a hard time with correctly reproducing structures with the same continuity as presented in the datasets.

Additionally, the global categorical facies distribution highlights a severe statistical imbalance that dictates the entire training dynamic. Reverting from the grouped three-class case—where the primary class dominates approximately 70% of the domain and the secondary appears only 5% of the time—to the high-fidelity case with nine distinct facies classes tests the ultimate operational limits of the conditional **GAN** framework. Managing nine simultaneous channels severely amplifies the structural complexity the network must resolve. The overwhelming volumetric presence of dominant sand bodies, such as *Point Bar* deposits, which occupy over 65% of the grid volume across all settings, stands in stark contrast to minority facies like the *Crevasse Splay Core*, *Crevasse Channel*, and *Crevasse Splay Delta*. These rare associations are suppressed down to occurrence rates between 0.0% and 0.1% due to fundamental boundary parameters. From a **ML** perspective this creates a critical optimization bottleneck.: because a standard **GAN** treats all voxels with equal weight, the discriminator is heavily incentivized to act as a shortcut detector, evaluating global sample quality predominantly on the dominant class. Consequently, while preliminary results from the high-fidelity model trained over 25 epochs indicate that large-scale structures maintain reasonable geological plausibility, the network struggles to accurately preserve minor facies. The low occurrence rates occasionally cause the model to find a stable loss minimum that completely discards these sparse elements. Crucially, the high-fidelity model struggles to achieve a good **MS-SWD** metric on these barely occurring classes. This poor statistical convergence is a direct result of the severe class imbalance and low occurrence rate, which overwhelms the mathematical contribution of these minority voxels to the global discriminator loss.

To mitigate these challenges in the primary configuration, the applied hyperparameter ablation study

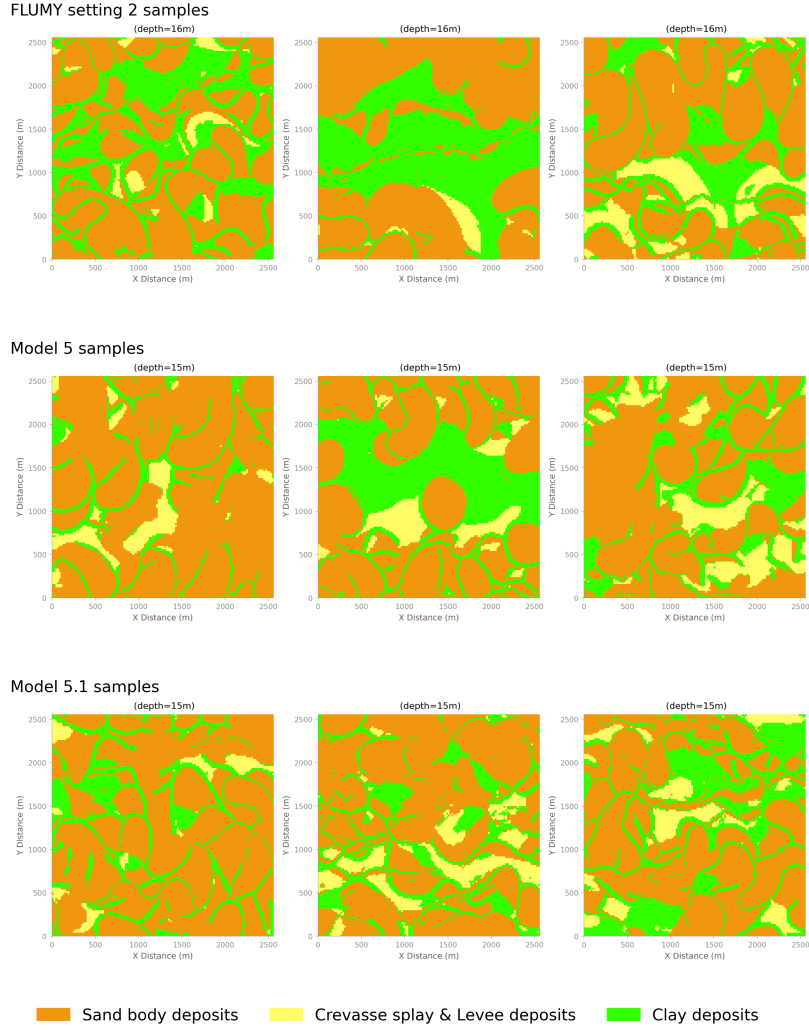


Figure 5.1: 2D slices taken from the Z axis at depth 16m of samples from 1) the *FLUMY* test dataset, 2) Model 5 (trained for 25 epochs) and 3) model 5.1 (trained for 100 epochs). ① denotes areas of closed loop structures whereas ② denotes areas with single isolated voxels.

proved essential for model stabilization. While base architectures suffered from mode collapse, achieving a normalized voxel-wise entropy score of just 0.515 compared to the baseline 0.69, stability was achieved by increasing network depth through double residual blocks and skip connections, which preserved high-frequency spatial gradients across the $128 \times 128 \times 32$ volume. Regularization was identified as the primary lever for equilibrium, with R1 Regularization applied every 16 iterations achieving a normalized voxel-wise entropy score of 0.704. Furthermore, the study highlights that model capacity is tightly coupled with data quantity; reducing the training volume by half caused significant degradation in the *MS-SWD* metric, confirming that higher-dimensional geological configurations require a sufficient dataset scale to ensure proper generalization.

Although the *MS-SWD* metric evolution provided no evidence of the class **balance** having an effect on the produced samples for the grouped classes, the connectivity analysis showed a clear mismatch in the blob size frequency of the two minority classes, Clay and Crevasse Splay & Levee deposits, compared to the test dataset (Figure 4.4). Moreover, well mismatch evaluation metrics (Figures 4.8 and 4.11) show that the minority classes are less accurate when it comes to adhering to hard well data. Crucially, this analysis uncovered a decoupling between automated metrics and true geological plausibility. While the *MS-SWD* metric plateaued after approximately 2,000 iterations, qualitative analysis indicated that topological connectivity—particularly for minority facies—continued to improve significantly with longer training. This suggests that while traditional distance metrics are effective for mon-

itoring statistical convergence, they are insufficient as standalone proxies for capturing the complex, long-range connectivity required in fluvial geomodelling.

In addition to the high-fidelity configuration, the evaluation of the multi-prior trained model revealed further generative limitations. Specifically, the multi-prior model failed to achieve good sample diversity, producing generated volumes with highly repetitive spatial configurations indicative of ensemble mode collapse. It is important to note, however, that no dedicated hyperparameter ablation study was performed on these multi-prior models. Given that stability in the base architecture was heavily reliant on rigorous tuning, it is highly probable that performing a similar hyperparameter study—adjusting network depth and regularization—could have mitigated this collapse and significantly improved the results.

Eventually, the following answers can be given to the sub-research questions defined earlier:

1. **How should the DSST's geology be parameterized in FLUMY to produce a training dataset sufficiently diverse for DGM modelling?**

The parameterization requires grouping the original nine high-fidelity facies into three broader classes to mitigate extreme volumetric imbalances and create a mathematically tractable prior. Additionally, the dataset must be generated at a sufficiently large scale to capture the necessary geological variance and prevent dataset-induced mode collapse.

2. **Which combination of hyperparameters yields a stable training process?**

Stability is primarily achieved by increasing network capacity through double residual blocks and skip connections to preserve high-frequency spatial gradients, combined with applying an R1 gradient penalty regularization every 16 iterations to maintain equilibrium.

3. **To what degree is the model able to reproduce the complexity inherent in the training data?**

While the model successfully mimics macro-scale volume fractions and large-scale horizontal and vertical dependencies (e.g., meandering channels), it struggles with the topological continuity of minority classes (e.g., broken mudplugs). Furthermore, it exhibits lower overall spatial entropy, indicating a tendency toward structural redundancy rather than exploring the full stochastic variance of the prior.

4. **To what degree does the model generate realisations that are successfully conditioned to well data?**

The two-step inversion process—combining latent space optimization with pivotal tuning—proved highly successful, achieving mean accuracy scores between 0.78 and 0.92 on hard vertical well constraints while preserving surrounding structural diversity, though conditioning accuracy remains lower for rare minority facies.

5. **How different do models trained on different geological priors react when sparse data is introduced?**

Models trained on multiple distinct geological priors struggled significantly, resulting in repetitive spatial configurations and ensemble mode collapse. This highlights the framework's hyperparameter fragility; a shift in the underlying geological prior requires a dedicated retuning of network depth and regularization to maintain generative stability.

5.2. Limitations of the current work

The process-based training datasets exhibit an extreme categorical disparity that introduces severe algorithmic challenges throughout the generative training phase. Dominant sand bodies, specifically Point Bar deposits, occupy over 65% of the global grid volume, while vital minority architectural elements—such as the Crevasse Splay Core and Crevasse Channel—are volumetrically suppressed to occurrence rates below 0.1%. Because the objective function of a standard GAN treats all geocellular voxels with uniform weight, the Discriminator is heavily incentivized to evaluate sample validity solely on the dominant class. Consequently, the Generator focuses almost exclusively on refining large-scale point bar structures, disregarding minority classes entirely. This statistical suppression is compounded by the framework's high susceptibility to ensemble mode collapse. While the model successfully mimics the macro-scale global volume fractions of the geological prior, spatial entropy metrics indicate a severe

drop, revealing that the network collapses into a narrow subset of stable configurations rather than exploring the full stochastic variance of the fluvial-deltaic system. For subsurface engineering, producing an ensemble with such artificially deflated spatial variance severely underestimates the true geological uncertainty, risking overly optimistic hydrodynamic simulation profiles. Furthermore, the training dynamics themselves exhibit significant hyperparameter fragility. The stability of the conditional GAN framework was heavily reliant on the specific ablation study performed for the Delft Geothermal Reservoir dataset, implying that transitioning this workflow to a different depositional environment would necessitate a complete and computationally expensive retuning of the network depth and regularization limits.

When transitioning the framework to real-world subsurface applications, significant physical scaling and data integration constraints emerge. The fixed spatial discretization of $128 \times 128 \times 32$ voxels strictly restricts the simulated domain to a localized volume of $2560 \text{ m} \times 2560 \text{ m} \times 32 \text{ m}$. While this resolution is mathematically sufficient to demonstrate a proof-of-concept inversion workflow between nearby wellbores, it cannot encapsulate regional-scale depositional sequences or extensive multi-scale heterogeneities without introducing severe truncation artifacts along the grid boundaries. A naive expansion of the 3D grid size to achieve full-field dimensions exponentially amplifies the memory footprint and training time, severely bottlenecking the current dual-GPU architecture. Additionally, the scope of the applied hard data conditioning presents a considerable operational limitation. The two-step inversion process—combining latent space optimization with pivotal tuning—proved successful for sparse, vertical well logs, but its efficacy when scaling to dense well networks, deviated and horizontal well trajectories, or integrating soft geophysical priors such as seismic data remains unproven within the scope of this study.

Evaluating the generated models also highlights a fundamental disconnect between standard automated ML metrics and true geological plausibility. While automated distance metrics such as the MS-SWD are highly effective for monitoring macro-scale statistical convergence, they often fail to accurately capture the topological validity of the generated structures. During training, the MS-SWD metric was observed to plateau relatively early, suggesting optimal convergence, yet qualitative and connectivity analyses indicated that the long-range continuity of minority facies continued to improve significantly with extended training. This discrepancy demonstrates that uniform statistical loss formulations and standard evaluation metrics are insufficient as standalone proxies for the complex, long-range connectivity required in dynamic fluid flow modeling.

As a final notable limitation, the standard convolutional architecture relies on an inherent assumption of spatial stationarity across the training patches. Natural fluvial-deltaic environments, however, exhibit distinct non-stationary trends, such as downstream fining or spatial variations in avulsion density, which the current unconstrained 3D GAN struggles to reproduce dynamically without the explicit integration of spatial conditioning trend maps.

5.3. Recommendations for Future Work

To further advance the application of DGM in real-world subsurface engineering, future research should address the architectural, mathematical, and physical limitations identified in this study. The most pressing recommendations are as follows:

- **Elimination of Topological Artifacts via Architectural Tuning:** As demonstrated, network depth, sufficient dataset scale, and rigorous hyperparameter combinations are essential for generative stability. Future work must investigate deeper or alternative convolutional architectures to fully resolve spatial continuity and eliminate the non-geological artifacts—such as closed-loop structures and broken mudplugs—observed in the current generated samples.
- **Transitioning to Continuous Grain Size Models:** To bypass the severe optimization bottlenecks caused by extreme categorical class imbalances, future workflows should evaluate modelling continuous properties (e.g., grain size or porosity) rather than discrete facies. Recent studies, such as those by [rongier2025](#) [towards](#) [empty citation](#), have shown highly promising results using continuous representations, which inherently smooth the mathematical gradients during adversarial training and may naturally preserve rare architectural elements.

- **Optimization of the Inversion Workflow:** While the two-step conditioning process successfully anchored realisations to hard well data, a dedicated hyperparameter ablation study focusing exclusively on the inversion phase is necessary. Systematically tuning the learning rates, the balance between well-mismatch loss and prior-preservation loss, and the pivotal tuning iterations could substantially improve the conditioning accuracy and reduce the frequency of non-honoring samples.
- **Mitigation of Class Imbalance via Advanced Inversion Loss Formulations:** If categorical facies modelling is retained, future iterations must abandon uniform voxel weighting during the data conditioning phase. Implementing a mathematically rigorous weighted Categorical Cross-Entropy (CCE) loss or a localized focal loss formulation for the context loss ($\mathcal{L}_{context}$) is strongly recommended. This will artificially amplify the gradients of vital, low-volume geobodies during latent space optimization and pivotal tuning, ensuring the inversion process does not strictly optimize for dominant classes while ignoring hard observations of minority facies at the wellbore.
- **Full-Field Upscaling and Spatial Scaling Constraints:** Research must expand beyond the strictly localized $128 \times 128 \times 32$ domain. To capture regional-scale depositional sequences without incurring exponential memory overhead, the workflow should transition to patch-based generation, progressive growing GANs, or multi-scale generative architectures ([federico2025three](#); [song2023gansim](#)).
- **Dynamic Fluid Flow and Geothermal Simulation:** Geothermal reservoir realisations that achieve high conditioning accuracy should be subjected to multi-phase dynamic fluid flow modelling. It is critical to quantify how the ML-generated topological artifacts and deflated spatial variance actually impact macroscopic reservoir behaviours, such as geothermal heat extraction rates and thermal breakthrough times, compared to traditional geostatistical baselines.

6 Conclusion

The application of GANs presents a promising, albeit highly complex, computational pathway for 3D geomodelling. By utilizing synthetic training datasets generated via the process-based software FLUMY, this research evaluated the capacity of a modified FluvGAN architecture to learn non-linear spatial relationships and replicate the fluvial-deltaic architectures of the DSST reservoir.

Addressing the central research question: “To what extent can GANs be used to create geologically plausible realisations of the DSST reservoir while conditioned to sparse real-world data?”, this study concludes that while GANs are able to generate geologically plausible, 3D reservoir realisations that honor real-world well data, so far they are not able to do both tasks with full accuracy. Moreover, the results achieved in this study were preceded by a long and tedious tasks of trying to find the right network and hyperparameter combination to let the model achieve geological plausibility as well as sample diversity. The model’s generative performance and conditioning capabilities also highlight several key outcomes and operational limitations:

- **Geological Plausibility and Generative Capacity:** The properly tuned 3-class GAN model successfully captured macro-scale 3D spatial dependencies, effectively replicating the dominant volumetric proportions and long-range structural features, such as meandering sand bodies. However, true geological plausibility is bottlenecked by the extreme categorical imbalances inherent in the training data. While dominant facies like point bars were reproduced with high confidence, the model struggled to accurately represent statistically suppressed minority facies (e.g., crevasse splays), occasionally resulting in topological artifacts such as broken mudplugs and isolated voxels.
- **Hyperparameter Fragility and Mode Collapse:** The model stability is incredibly sensitive to hyperparameter configurations. Expanding the model’s scope to handle nine individual facies classes or a multi-prior dataset spanning varied depositional rules triggered severe optimization bottlenecks and ensemble mode collapse.
- **Conditioning to Sparse Real-World Data:** Despite generative challenges, the two-step inversion workflow (latent space optimization combined with pivotal tuning) proved highly effective in anchoring the synthetic realisations to hard field observations. When evaluated against the DEL-GT-01 and DEL-GT-02-S2 well trajectories, the conditioned realisations achieved strong localized accuracy scores ranging from 0.78 to 0.92. The generative ensemble retained a reasonable degree of stochastic diversity in the unobserved regions, although expanding the radial constraint threshold around the wellbores naturally diminished broader spatial entropy.

Ultimately, while GANs offer a computationally efficient method for generating vast, conditioned ensembles suitable for uncertainty quantification, transitioning this framework to full-field subsurface applications will require advanced loss formulations and architectural upscaling to resolve minority topological features without sacrificing stochastic variance.

6.1. Final Remarks

A Delft Geothermal Reservoir well data

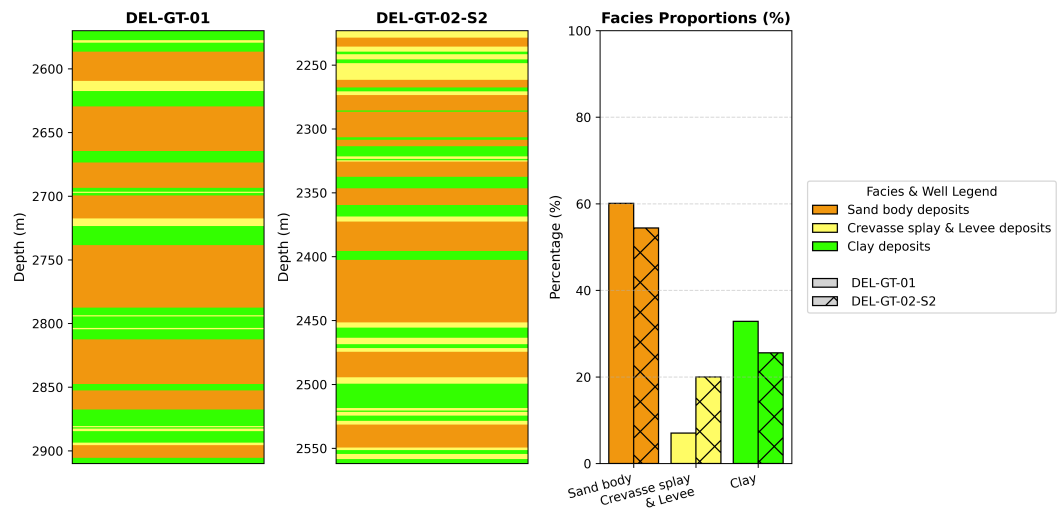


Figure A.1: Overview of FAs by **(bruining2023sedimentological)** on the DEL-GT-01 and DEL-GT-02-S2 reservoir section. The Y-axis represents the along hole depth.

B FLUMY dataset generation

Value	Facies Key	Color	Description
0	Background		Undefined
1	Channel Lag		Coarse active channel fill
2	Point Bar		Meandering lateral accretion
3	Sand Plug		Fine-grained channel abandonment
4	Crevasse Splay I		Proximal high-energy overbank
5	Crevasse Channel		Feeder for splay deposits
6	Crevasse Splay II		Feeder for splay deposits
7	Levee		Bordering channel ridges
8	Overbank		Vegetated floodplain silt
9	Mud plug		Organic-rich anoxic sediments
10	Hemipelagic plug		Fine-grained marine deposits consisting of clay and silt-sized grains
11	Wetland/Paleosol		Organic-rich anoxic sediments
12	Draping		Fine-grained sediments conforming to topography
13	Pelagic		Deep-marine fine-grained suspension settling

Table B.1: Overview of the 9 facies classes that populated the training dataset. Due to specific setting in the flumy default non expert user settings (see figure 3.1), only 9 facies populated the eventual datasets.

Lithofacies identifiers and their default grain size class:														
Lithofacies ID	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Lithofacies	Pelagic	Draping	Wetland	Hemipela. Plug	Mud Plug	Overbank	Levee	Crevasse Splay II	Crev. Spl. Channel	Crevasse Splay I	Sand Plug	Point Bar / LAPS	Channel Lag	Undefined
Abbreviation	PL	DR	WL	HP	MP	OB	LV	CS2	CCH	CS1	SP	PB	CL	UDF
Default Grain Size Class	1	1	1	2	2	3	6	7	8	9	9	10	13	0
Default ϕ Class	11:14	11:14	11:14	9:10	9:10	8	5	4	3	2	2	1	-2	NA

Grain size classes and lithofacies ranges:															
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Undefined	Clay (11:14)	Clay (9:10)	Clay (8)	Silt (7)	Silt (6)	Silt (5)	Silt (4)	Very Fine Sand (3)	Fine Sand (2)	Medium Sand (1)	Coarse Sand (0)	Very Coarse Sand (-1)	Gravel (-2)	Pebble (-3:-5)	Cobble (-6:-8)
0.0625	0.125	0.1875	0.25	0.3125	0.375	0.4375	0.5	0.5625	0.625	0.6875	0.75	0.8125	0.875	0.9375	1
	Pelagic Draping Wetland	Mud Plug Hemipelagic Plug						Crevasse Splay II		Crevasse Splay Channel					
		Overbank					Levee			Crevasse Splay I Point Bar / LAPS				Channel Lag	
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Figure B.1: Overview of all possible facies defined in the flumy user guide (add citation) with the respecting range of grain sizes.

C FluvGAN

C.1. One Hot Encoding Strategy

The one-hot encoding strategy used in this project created n spatial channels based on the number of individual classes passed on to the model. Where traditional one-hot encoding strategies map the values in a channel to either 1 or 0, referring to the presence or absence of a specific class, the final *Tanh* activation function in the **FluvGAN** model maps values to either -1 or 1. Figure C.1 provides an overview of one hot encoding of a vector. Using one-hot encoding categorical independent data can be

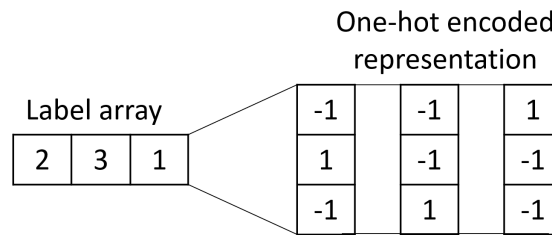


Figure C.1: Simple one-hot encoding representation by converting the original data array into a multitude of bipolar vectors. One-hot encoding can be applied to any type of categorical data.

correctly fed to the model, avoiding issues where the model thinks the numbers/values are correlated. The model may start to correlate the classes based on values, tending to overestimating the classes that do not occur often.

C.2. Fluvgan Architecture

Layer / Block Type	Output Shape	Param #
ConvBlockTranspose3d (Scale 1)	[1, 1024, 4, 4, 4]	6,553,600
DeepResBlockUpscale3d (Scale 2)	[1, 1024, 8, 8, 8]	4,066,816
DeepResBlock3d (Scale 2)	[1, 1024, 8, 8, 8]	4,066,816
DeepResBlockUpscale3d (Scale 3)	[1, 512, 16, 16, 16]	3,935,744
DeepResBlock3d (Scale 3)	[1, 512, 16, 16, 16]	1,017,600
DeepResBlockUpscale3d (Scale 4)	[1, 256, 32, 32, 32]	984,832
DeepResBlock3d (Scale 4)	[1, 256, 32, 32, 32]	254,848
DeepResBlockUpscale3d (Scale 5)	[1, 128, 32, 64, 64]	99,200
DeepResBlock3d (Scale 5)	[1, 128, 32, 64, 64]	27,072
DeepResBlockUpscale3d (Scale 6)	[1, 64, 32, 128, 128]	25,024
DeepResBlock3d (Scale 6)	[1, 64, 32, 128, 128]	6,880
BatchNorm3d + LeakyReLU	[1, 64, 32, 128, 128]	128
ConvBlock3d (Output)	[1, 9, 32, 128, 128]	15,552
Total Trainable Parameters		21,054,112

Table C.1: Detailed architectural specifications and layer parameters for the DeepGenerator3d network.

Layer / Block Type	Output Shape	Param #
ConvBlock3d (Input)	[1, 64, 32, 128, 128]	15,552
DeepResBlockDownscale3d (Scale 2)	[1, 128, 16, 64, 64]	20,992
DeepResBlock3d (Scale 2)	[1, 128, 16, 64, 64]	63,488
DeepResBlockDownscale3d (Scale 3)	[1, 256, 8, 32, 32]	83,968
DeepResBlock3d (Scale 3)	[1, 256, 8, 32, 32]	253,952
DeepResBlockDownscale3d (Scale 4)	[1, 512, 4, 16, 16]	335,872
DeepResBlock3d (Scale 4)	[1, 512, 4, 16, 16]	1,015,808
DeepResBlockDownscale3d (Scale 5)	[1, 1024, 4, 8, 8]	753,664
DeepResBlock3d (Scale 5)	[1, 1024, 4, 8, 8]	1,703,936
DeepResBlockDownscale3d (Scale 6)	[1, 1024, 4, 4, 4]	1,703,936
DeepResBlock3d (Scale 6)	[1, 1024, 4, 4, 4]	1,703,936
LeakyReLU	[1, 1024, 4, 4, 4]	0
LinearBlock (Output Prediction)	[1, 1]	1,025
Total Trainable Parameters	7,656,129	

Table C.2: Detailed architectural specifications and layer parameters for the DeepDiscriminator3d network.

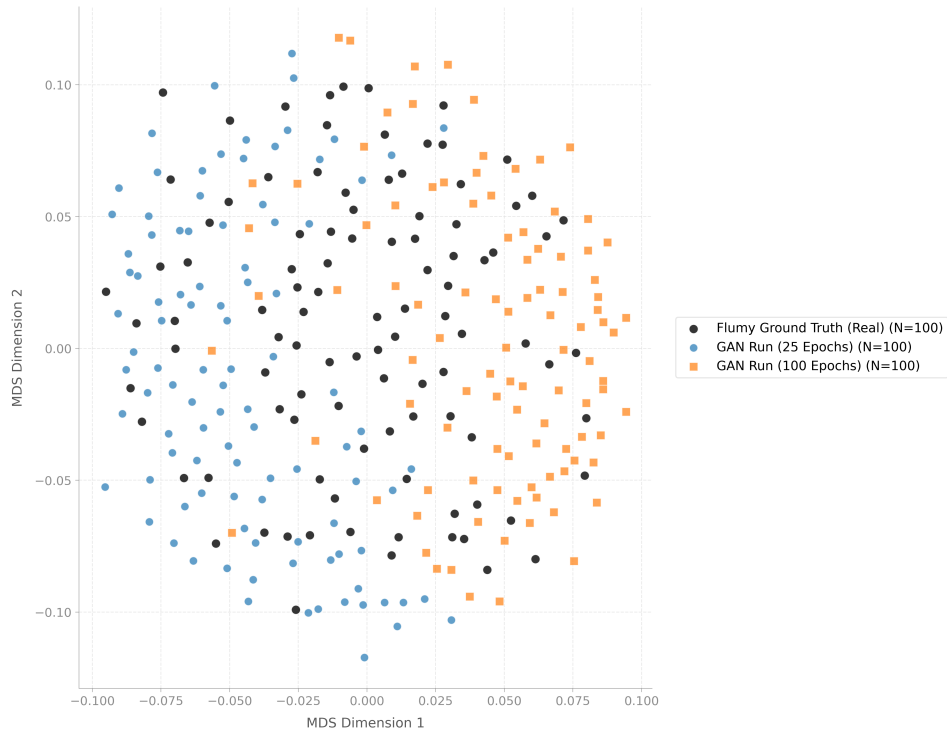


Figure C.2: MDS plot of the MS-SWD values computed for the test dataset, illustrating the distribution shift and clustering behaviour of the model 5 architecture between 25 and 100 training epochs.

D Mathematical Formulations for Optimization and Evaluation

D.1. Optimization & Inversion Framework

The two-step inversion process for conditioning the 3-facies generative models is mathematically defined by consecutive optimization phases: Latent Space Optimization followed by Pivotal Tuning.

D.1.1. Latent Space Optimization

During the initial Latent Space Optimization, the internal weights of the generator G are frozen. Gradient descent is utilized to iteratively search the latent space $\mathcal{Z}$ for an optimal latent noise vector $\mathbf{z}^*$ that minimizes the loss between the generated facies and the true geological observations.

Rather than evaluating the penalty exclusively at the exact well coordinates, a spatially weighted context mask is applied to enforce localized geological alignment without over-constraining the broader, unobserved reservoir architecture. A spatial influence radius is established around each well point, governed by a distance threshold (σ). This threshold acts as the standard deviation for a continuous Gaussian decay function, dictating the geostatistical “reach” of the well data.

Because fluvial-deltaic depositional systems are highly anisotropic—characterized by extensive horizontal continuity and limited vertical thickness—an anisotropic distance metric $d_A(v)$ is utilized, penalizing vertical deviations significantly more than horizontal ones. For any voxel v in the 3D grid $\mathcal{V}$, the spatial weight mask $M(v)$ is defined as:

$$M(v) = \begin{cases} \exp\left(-\frac{d_A(v)^2}{2\sigma^2}\right), & \text{if } d_A(v) \leq 3\sigma \\ 0, & \text{if } d_A(v) > 3\sigma \end{cases} \quad (\text{D.1})$$

While the threshold (σ) controls the smooth, continuous spread of the penalty, calculating gradients for distant voxels is computationally inefficient. Therefore, the mask is explicitly truncated to zero at a hard computational cutoff of 3σ , where the gradient contribution becomes mathematically negligible.

The objective function is thus defined as a spatially masked Context Loss, replacing the point-wise evaluation. It is evaluated over the grid volume $\mathcal{V}$ and the C total facies classes:

$$\mathbf{z}^* = \arg \min_{\mathbf{z}} \left[\sum_{v \in \mathcal{V}} \sum_{i=1}^C M(v) \cdot \mathcal{L}_{CE}(p_{v,i}, y_{v,i}) \right] \quad (\text{D.2})$$

where $\mathcal{L}_{CE}$ is the Categorical Cross-Entropy. Specifically, $y_{v,i}$ is the binary truth value (0 or 1) indicating whether class i is the correct broadcasted facies for voxel v (based on the nearest well data point), and $p_{v,i}$ is the generator’s predicted probability for that class.

Computational Scaling: Latent optimization is highly efficient. The computational cost scales linearly, $\mathcal{O}(N_w \cdot C)$, with the number of conditioning points N_w and facies classes C . Because the network weights are frozen, the gradient is only computed with respect to the latent vector $\mathbf{z}$ (typically of dimension 512 or 1024), making this phase fast and highly scalable to dense well configurations.

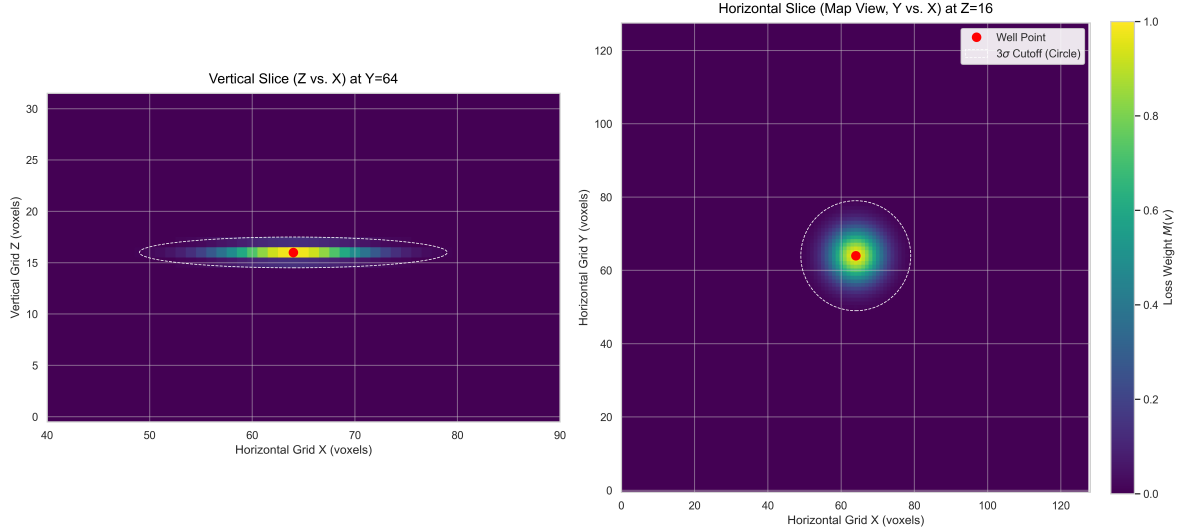


Figure D.1: Orthogonal cross-sections illustrating the 3D anisotropic Gaussian weight mask $M(v)$ applied during Latent Space Optimization. The anisotropy factors are 10,1,1. The distance threshold (σ) controls the smooth spatial decay of the penalty away from the well point (red dot). Due to the anisotropic scaling factor, the mask forms a horizontal lens in the vertical cross-section (left), respecting stratigraphic deposition, while maintaining a perfectly radial spread in the map view (right). The gradient is safely truncated at a 3σ limit (dashed line) to preserve computational efficiency.

D.1.2. Pivotal Tuning

To overcome the inherent boundaries of the learned latent space, Pivotal Tuning (**roich2022pivotal**) is applied. Using the previously optimized $\mathbf{z}^*$ as a static anchor, the generator’s internal weights θ are unfrozen and optimized:

$$\theta^* = \arg \min_{\theta} \mathcal{L}_{CE}(G_{\theta}(\mathbf{z}^*)_{\text{well}}, \mathbf{y}_{\text{well}}) + \lambda \|\theta - \theta_0\|_2 \quad (\text{D.3})$$

This formulation pushes the localized outputs to perfectly match the hard conditioning data. The critical addition here is the L_2 regularization term $\lambda \|\theta - \theta_0\|_2$, where θ_0 represents the original pre-trained weights and λ is a penalty coefficient. This term forces the modified weights to stay as close as possible to their original state, preserving the global geological realism of the model and preventing catastrophic forgetting.

Computational Scaling: Unlike latent optimization, Pivotal Tuning is computationally intensive. The optimization scales with the total number of trainable parameters in the generator, $|\theta|$, which often numbers in the tens of millions for 3D generative networks, $\mathcal{O}(|\theta| \cdot N_w)$. It requires backpropagation through the entire network architecture for every iteration, scaling linearly with the size of the 3D grid output during the forward pass.

D.2. Generative Evaluation Metrics

The evaluation framework relies on advanced statistical divergence and optimal transport metrics to overcome the limitations of standard voxel-by-voxel assessments in unconditioned 3D modeling.

D.2.1. Wasserstein Distance and Mean SWD

The conceptual foundation for structural evaluation is the Wasserstein distance, which quantifies the optimal transport cost required to reconfigure the generated probability distribution P_g into the reference real data distribution P_r (**panaretos2019statistical**; **rongier2025atowards**). It is defined as:

$$\mathcal{W}(P_r, P_g) = \inf_{\gamma \sim \Pi(P_r, P_g)} \mathbb{E}_{(x,y) \sim \gamma} [\|x - y\|] \quad (\text{D.4})$$

where $\Pi(P_r, P_g)$ denotes the set of all joint probability distributions $\gamma(x, y)$ whose marginals are P_r and P_g respectively.

Because computing Equation D.4 over high-dimensional 3D grids suffers from the curse of dimensionality, we employ the Mean **SWD** (MSWD). MSWD projects the complex 3D distributions onto 1D lines where the optimal transport problem has a trivial, closed-form solution (sorting). Mathematically, it is defined by integrating the 1D Wasserstein distances over the unit hypersphere $\mathbb{S}^{d-1}$:

$$\text{MSWD}(P_r, P_g) = \int_{\mathbb{S}^{d-1}} \mathcal{W}_{1D}(\mathcal{R}_\omega \# P_r, \mathcal{R}_\omega \# P_g) d\omega \quad (\text{D.5})$$

where $\mathcal{R}_\omega \# P$ denotes the projection (Radon transform) of distribution P onto a 1D line defined by the direction vector ω . In practice, this integral is approximated using a Monte Carlo approach by averaging over a large number of random projection angles L .

Computational Scaling: Computing the exact Wasserstein distance scales at roughly $\mathcal{O}(V^3 \log V)$, where V is the total number of voxels in the 3D grid. This is intractable for standard geomodels. MSWD circumvents this by scaling at $\mathcal{O}(L \cdot V \log V)$, where L is the number of projection slices. Since sorting 1D arrays is exceptionally fast, MSWD allows for robust structural comparison of massive 3D grids in a fraction of the time.

D.2.2. Uncertainty and Spatial Divergence Metrics

To assess spatial uncertainty and validate generation diversity across an ensemble, Normalized Shannon's Entropy (**shannon1948mathematical**) is calculated independently for every spatial coordinate (voxel) $\mathbf{v} \in \mathbb{R}^3$. It scales the standard entropy equation by the number of geological classes (k) to bound the metric between 0 and 1:

$$H_{\text{norm}}(\mathbf{v}) = -\frac{1}{\log_2(k)} \sum_{i=1}^k P(F_i|\mathbf{v}) \log_2 P(F_i|\mathbf{v}) \quad (\text{D.6})$$

Here, $P(F_i|\mathbf{v})$ is the empirical probability of generating facies F_i at the specific voxel $\mathbf{v}$ across the entire ensemble of realizations. An $H_{\text{norm}}(\mathbf{v})$ of 0 implies total certainty (uniformity or potential mode collapse at that pixel), while 1 implies total uncertainty (maximum diversity across all k facies).

Computational Scaling: Entropy calculations are highly efficient parallel operations. The scaling is purely linear, $\mathcal{O}(V \cdot k)$, where V is the total number of voxels, making it perfectly suited for massive 3D grids regardless of ensemble size.

To quantify the distributional disagreement between the true reference models (P) and the generated models (Q), we calculate the pixel-wise Kullback-Leibler (KL) divergence (**kullback1951information**):

$$D_{KL}(P_{\mathbf{v}} \parallel Q_{\mathbf{v}}) = \sum_{i=1}^k P(F_i|\mathbf{v}) \log_2 \left(\frac{P(F_i|\mathbf{v})}{Q(F_i|\mathbf{v})} \right) \quad (\text{D.7})$$

Because KL divergence is asymmetric and unbounded, it is transformed into the Jensen-Shannon (JS) divergence (**lin1991divergence**) to provide a bounded, symmetric measure of stochastic spread at each voxel:

$$D_{JS}(\mathbf{v}) = \frac{1}{2} D_{KL}(P_{\mathbf{v}} \parallel M_{\mathbf{v}}) + \frac{1}{2} D_{KL}(Q_{\mathbf{v}} \parallel M_{\mathbf{v}}) \quad (\text{D.8})$$

where $M_{\mathbf{v}} = \frac{1}{2}(P_{\mathbf{v}} + Q_{\mathbf{v}})$ is the mixture distribution at voxel $\mathbf{v}$.

Finally, to aggregate this 3D grid of spatial conflict into a single definitive scalar for model evaluation, the mean JS divergence is computed across the entire spatial volume $\mathcal{V}$:

$$\overline{D}_{JS} = \frac{1}{|\mathcal{V}|} \sum_{\mathbf{v} \in \mathcal{V}} D_{JS}(\mathbf{v}) \quad (\text{D.9})$$

Computational Scaling: Similar to entropy, JS Divergence operates strictly on the pre-computed probability mass functions of the respective ensembles. Therefore, the scalar aggregation scales linearly with the number of voxels evaluated, $\mathcal{O}(V)$. It remains computationally trivial even for massive, high-resolution spatial domains.

To aggregate this spatial conflict into a single quantitative scalar comparing two datasets, we utilize the Jensen-Shannon (JS) divergence (**lin1991divergence**). It measures the symmetric stochastic spread between the true reference distributions (P) and the generated distributions (Q):

$$D_{JS}(P \parallel Q) = \frac{1}{2}D_{KL}(P \parallel M) + \frac{1}{2}D_{KL}(Q \parallel M) \quad (\text{D.10})$$

where $M = \frac{1}{2}(P + Q)$ is a mixture distribution, and D_{KL} is the standard Kullback-Leibler divergence (**kullback1951information**), explicitly formulated as:

$$D_{KL}(P \parallel Q) = \sum_x P(x) \log \left(\frac{P(x)}{Q(x)} \right) \quad (\text{D.11})$$

Unlike D_{KL} , which is asymmetric and can approach infinity if $Q(x)$ is zero, D_{JS} is strictly symmetric ($D_{JS}(P \parallel Q) = D_{JS}(Q \parallel P)$) and bounded between $[0, 1]$ when calculated with base-2 logarithms, making it a robust final scalar for spatial fidelity.

Computational Scaling: Similar to entropy, JS Divergence operates strictly on the pre-computed probability mass functions of the respective ensembles. Therefore, it scales linearly with the number of spatial bins/voxels evaluated, $\mathcal{O}(V)$. It remains computationally trivial even for massive, high-resolution spatial domains.

E Evaluation Metrics

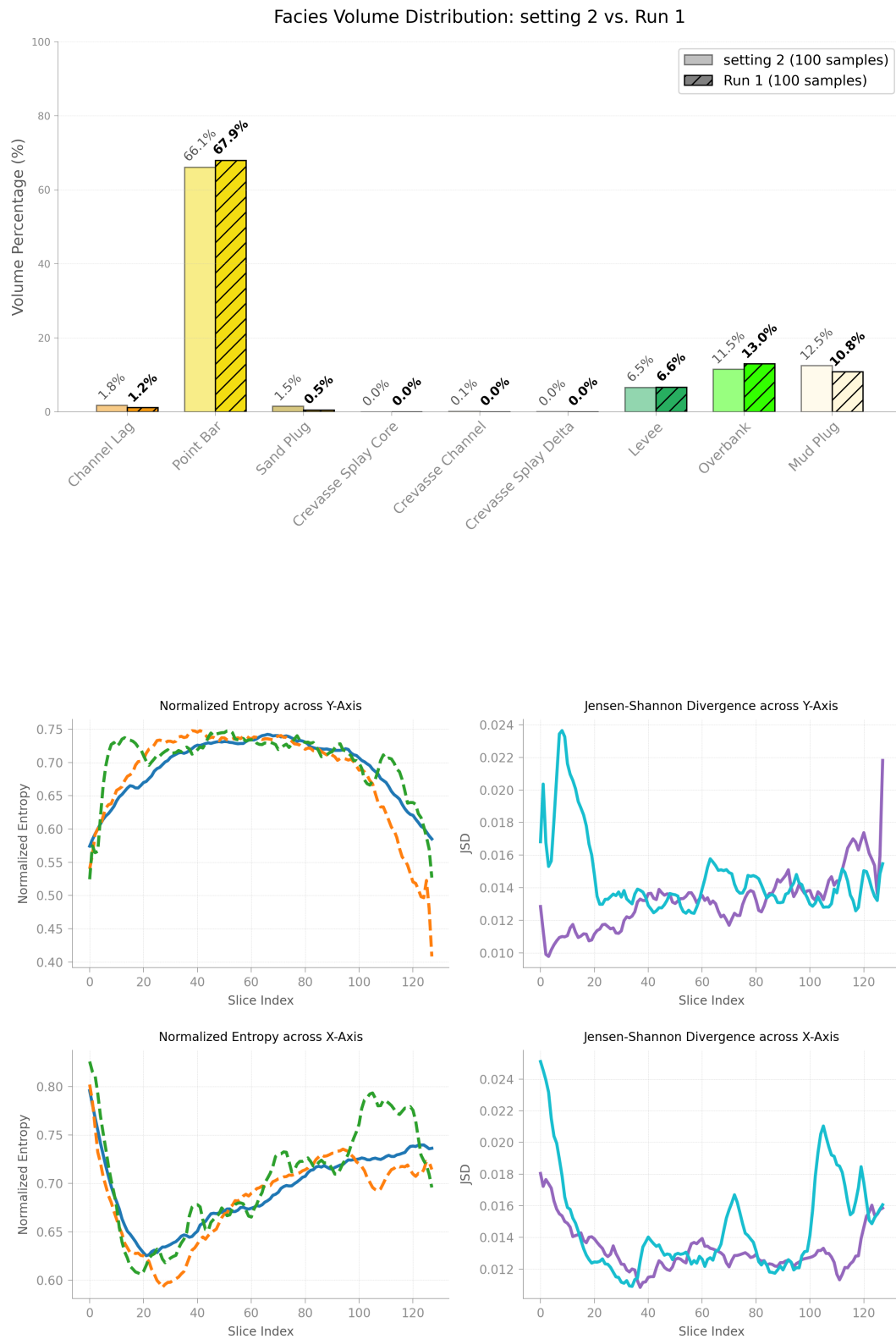
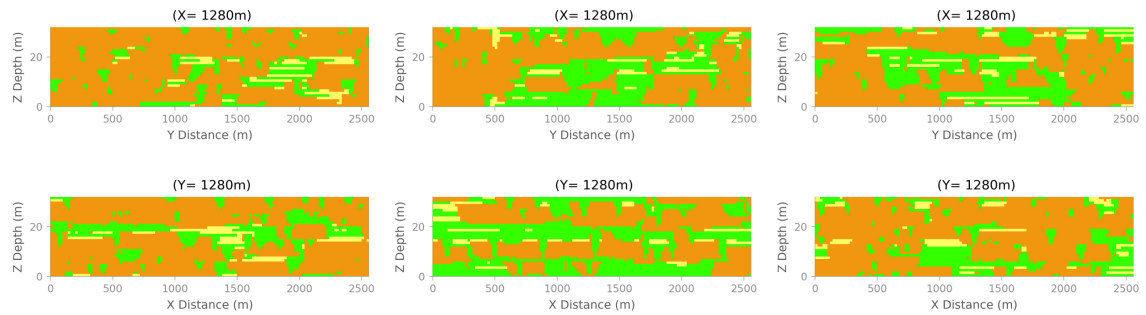
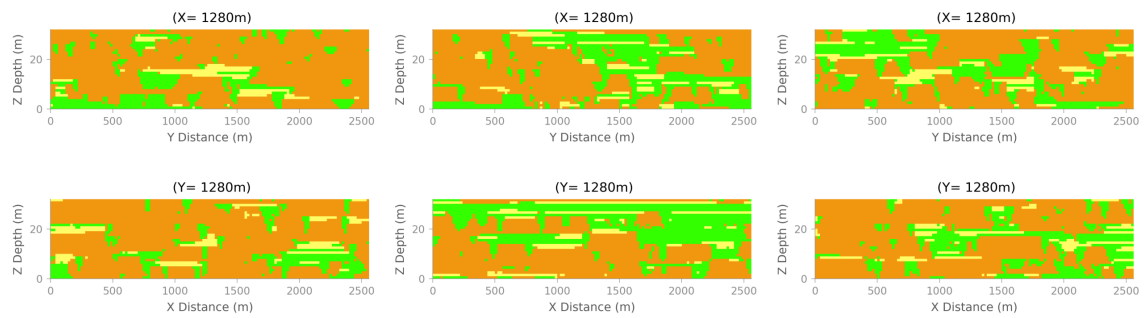


Figure E.1: Comparison of the JSD normalized voxel-wise entropy across the Y, and X axes.

FLUMY setting 2 samples (X- and Y- Axis Sections)



Model 5 samples (X- and Y- Axis Sections)



Model 5.1 samples (X- and Y- Axis Sections)

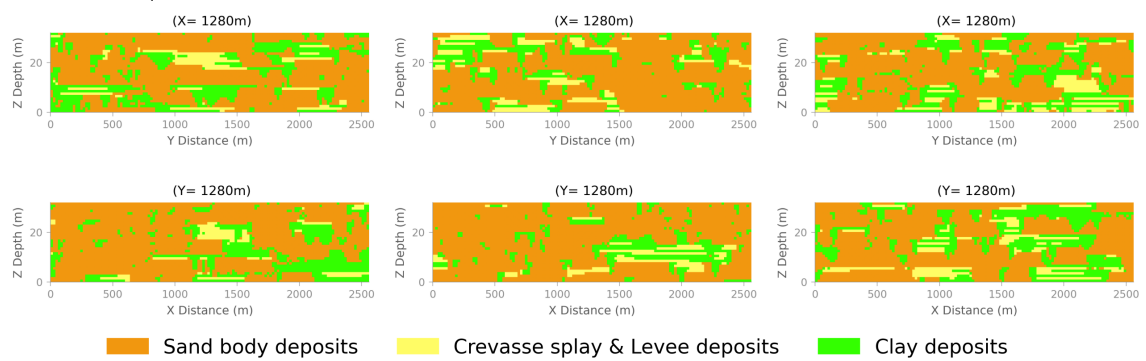


Figure E.2: tba